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TACKLING FALSE POSITIVES IN BUSINESS RESEARCH: A STATISTICAL TOOLBOX WITH APPLICATIONS

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Abstract. Serious concerns have been raised that false positive findings are widespread in empirical research in business disciplines. This is largely because researchers almost exclusively adopt the '*p*-value less than 0.05' criterion for statistical significance; and they are often not fully aware of large-sample biases which can potentially mislead their research outcomes. This paper proposes that a statistical toolbox (rather than a single hammer) be used in empirical research, which offers researchers a range of statistical instruments, including a range of alternatives to the *p*-value criterion such as the Bayesian methods, optimal significance level, sample size selection, equivalence testing and exploratory data analyses. It is found that the positive results obtained under the *p*-value criterion cannot stand, when the toolbox is applied to three notable studies in finance.

Keywords. Adaptive significance level; Bayes factor; Decision theory; Fallacy of rejection; Large-sample bias; Zero probability paradox

JEL Classification C12; G10

If all you have is a hammer, everything looks like a nail.

Law of the Instrument
Abraham Maslow, 1966
Warren Buffet, 1984

1. Introduction

There are growing concerns and accumulated evidence that most scientific findings are the outcomes of false positives (see, for example, Innoniades, 2005; Rao and Lovric, 2016; Wasserstein and Lazar, 2016). In finance, Kim and Ji (2015) provide such empirical evidence, while Harvey *et al.* (2016) and Harvey (2017) are concerned with the false positives resulting from multiple testing and *p*-hacking. In accounting, Basu (2015), Dyckman (2016) and Ohlson (2015, 2018), among others, voice similar concerns, while Kim *et al.* (2018) provide the empirical evidence. Regarding other areas of science, the authors who raise the concerns include Keuzenkamp and Magnus (1995) and McCloskey and Ziliak (1996) for economics; Lin *et al.* (2013) for information systems; Lockett *et al.* (2014) for management; Cumming (2013) for

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psychology; and Sawyer and Peter (1983) and Hubbard and Armstrong (2006) for marketing. Moosa (2017) also provides a comprehensive critique on the practice of econometrics, identifying the elements of significance testing (p-hacking and an arbitrary choice of significance level) as the shortcomings of econometric analysis. At the very heart of the problem is what Gigerenzer (2004) called the 'null ritual', in which a statistical decision for a sharp or point null hypothesis is made almost exclusively employing the 'p-value less than α' criterion, where α denotes the level of significance. While the p-value can be a highly misleading measure of evidence for a sharp null hypothesis (Berger and Sellke, 1987; Startz, 2014), the threshold of decision α is often chosen arbitrarily (at 0.05, 0.01 or 0.10) with little scientific justification (Arrow, 1960, p.70; Lehmann and Romano, 2005, p.57; Kim and Ji, 2015). In addition, the power of the test (or the probability of Type II error) is often totally ignored in such decision making (Hausman, 1978; Leamer, 1978; MacKinnon, 2002; Startz, 2014; Kim and Ji, 2015). The latter is particularly problematic for business studies, where large or massive samples are routinely adopted. In this case, the power is often practically one and the probability of Type II error is 0, rendering the statistical test severely biased towards Type I error if the value of α is set at a conventional value such as 0.05, with the consequence of frequently rejecting true null hypotheses (see, for example, Kim, 2017b).

It also appears that many empirical researchers are overconfident about the quality of their data, when they have large or massive sample sizes (often called 'big data hubris'). A number of authors from many areas of science raise the point that large data sets often suffer from a range of problems (sampling error, selection bias, measurement error, missing information, aggregation error, etc.), which can severely bias research outcomes and deliver poor decisions (see, for example, Boyd and Crawford, 2011; Wigan and Clarke, 2013; Harford, 2014; Hand, 2016, 2018; and Fiebig, 2017). More problematic is that these biases can be magnified with increasing sample size, as Kaplan *et al.* (2014) point out. This means that the researchers using large data sets should conduct extensive cautionary analyses, which require attention to 'small data sets' (Hand, 2016) or 'sample splitting' (Ohlson, 2015), along with appropriate sensitivity analyses to address potential large-sample biases (Kaplan *et al.*, 2016).

In this paper, I propose that researchers employ a range of alternatives for their significance testing for more sound and sensible statistical decisions, along with cautionary analyses to identify the potential large-sample bias. This is partly in pursuit of Gigerenzer's (2004) proposal that the researchers employ a statistical toolbox (containing a range of alternatives) rather than a single hammer (the p-value criterion). It also follows the recommendations made by recent critiques of the p-value criterion that researchers consider other alternatives, including the Bayesian method, a tighter threshold for statistical significance and estimation-based method such as confidence interval (see, for example, Cumming, 2013; Wasserstein and Lazar, 2016; Benjamin *et al.*, 2017; and Harvey, 2017). This is also in light of the warnings against the overconfidence of the researchers for large sample and possible large-sample bias in estimation and inference (Harford, 2014; Kaplan *et al.*, 2014; Hand, 2016; Fiebig, 2017). By adopting such a toolbox in empirical research, we can: firstly, promote statistical thinking in our decision making; and secondly, avoid 'mindless statistics' and 'statistical rituals' that can lead to false positive research outcomes (Gigerenzer, 2004; Ohlson, 2018).

The purpose of this paper is to introduce such a statistical toolbox for empirical researchers, and to apply it to three notable published studies in finance. The instruments in the toolbox include the optimal level of significance based on a decision-theoretic approach (Leamer, 1978), adaptive level of significance (Perez and Pericchi, 2014), Bayes factor (Zellner and Siow, 1980; Liang *et al.*, 2008), equal-probability test (Arrow, 1960), approximate probability of the null hypothesis being true (Startz, 2014) and equivalence testing (Lauzon and Caffo, 2009; Rao and Lovric, 2016). The toolbox also includes cautionary data analyses for possible large-sample bias estimation and inference, when the researcher has a large or massive data set. Complemented with the popular p-value criterion, these alternatives can help researchers avoid research outcomes that are false positives. The toolbox is used to re-evaluate the results of three important studies in empirical finance. It is found that the positive results obtained under the p-value criterion cannot generally stand when the toolbox is applied to these studies. In the next section, I present the background, including the problems associated with the *p*-value criterion. Section 3 presents the details of the proposed statistical toolbox; Section 4 presents the empirical applications; and Section 5 concludes the paper.

2. Background

This section provides the details of the model and statistical test. For simplicity, the details are presented in the context of a multiple regression, which includes a wide range of linear models including the fixed-effects model. The discussion also includes the zero probability paradox (Rao and Lovric, 2016); and the shortcomings of the *p*-value criterion, especially under large or massive sample sizes.

2.1 *Basics*

For the purpose of exposition, consider a standard linear regression model

$$Y = \gamma_0 + \gamma_1 X_1 + \cdots + \gamma_K X_K + u \tag{1}$$

where Y is the dependent variable, Xs are independent variables and γs are unknown coefficients, and u is the error term following an independent normal distribution with a fixed variance. To test for $H_0 : \gamma_1 = \cdots = \gamma_J = 0$ $(J \leq K)$ against $H_1 : \gamma_j \neq 0$ (for at least one value of j), we use the F-test whose statistic can be expressed as:

$$F = \frac{\left(R_1^2 - R_0^2\right)/J}{\left(1 - R_1^2\right)/(T - K - 1)} \tag{2}$$

where R_j^2 represents the coefficient of determination under H_j $(j = 0, 1)$ and T is the sample size. Under the assumption of normality, the F statistics follows the (central) F-distribution with J and $T - K - 1$ degrees of freedom, denoted as $F(J, T - K - 1)$. Under H_1, the F-statistic follows a non-central F-distribution $F(J, T - K - 1; \lambda)$, where λ denotes the non-centrality parameter, which is written as

$$\lambda = T \frac{R_{p1}^2 - R_{p0}^2}{1 - R_{p1}^2} \tag{3}$$

where R_{pj}^2 denotes the population coefficient of determination under H_j (Poirier, 1995, theorem 9.6.1; Peracchi, 2001, theorem 9.2). In the simple linear regression model where X_1 is the only explanatory variable,

$$R_{p1}^2 = \gamma_1^2 \frac{Var(X_1)}{Var(Y)} \tag{4}$$

and $R_{p0}^2 = 0$ when $H_0 : \gamma_1 = 0$. Note from (4) that R_{p1}^2 is directly related with the effect size γ_1: a low value of R_{p1}^2 implies a small effect size, given the variance ratio of X_1 to Y. A similar relationship to (4) also holds for a general case of $K > 1$, without loss of generality. Let

$$\eta \equiv \frac{R_{p1}^2 - R_{p0}^2}{1 - R_{p1}^2} \tag{5}$$

which may be called the incremental signal-to-noise ratio of H_1 to H_0, with its sample estimator denoted as $\hat{\eta} = \frac{R_1^2 - R_0^2}{1 - R_1^2}$. Since $F = \hat{\eta}(T - K - 1)/J$ and $\lambda = T\eta$, it is clear that the F-statistic and its distribution under H_1 respond sharply to sample size T. These expressions show how statistical inference is related with model fit (signal-to-noise ratio) and sample size, although the latter plays a much more dominant role. Note that similar results hold for the *t*-test on a single regression coefficient in an equivalent manner with $t = \sqrt{F}$.

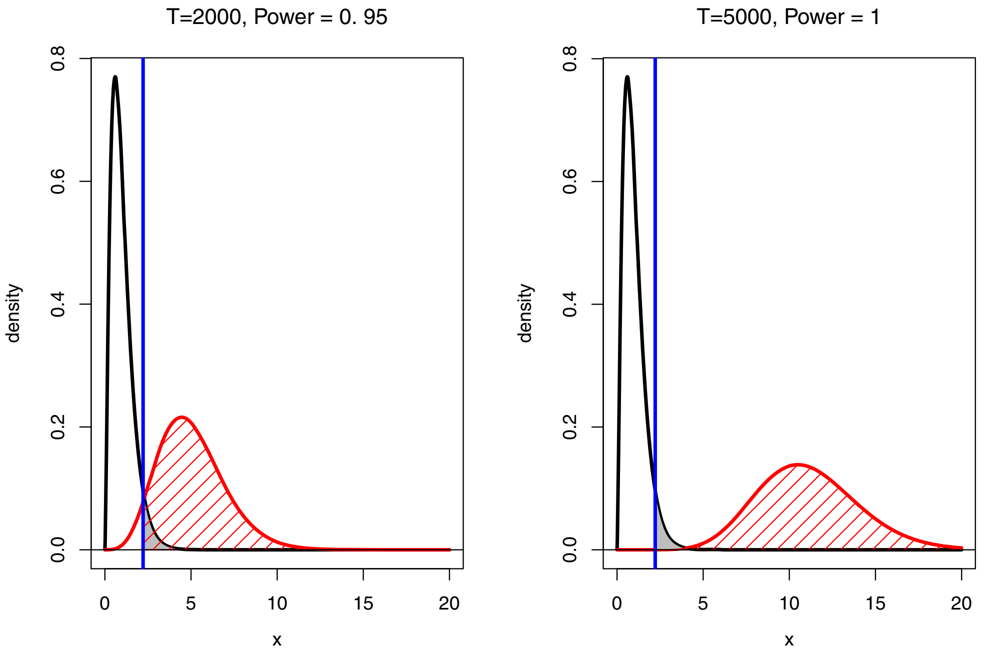

Figure 1. Distributions under H_0 and H_1 for the F-Test. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The black curve plots the density of F-statistic against its support (labelled x) under $H_0 : \gamma_1 = \cdots = \gamma_5 = 0$. The red curve plots the density of F-statistic under $H_1 : \gamma_j \neq 0$: it is associated with the value of λ given in (3), when $R^2_{p1} = 0.01$, $R^2_{p0} = 0$ and $K = J = 5$. The blue vertical line indicates the critical value of the F-test at $\alpha = 0.05$ (approximately 2.22). The grey shaded area corresponds to the level of significance (Type I error probability), while the red shaded area indicates the power $(1 - \beta)$ of the test which is the probability of rejecting H_0 when it is false.

Figure 1 plots the density functions of the F-statistic given in (2) under two different values of T, when $R^2_{p1} = 0.01$, $R^2_{p0} = 0$, $J = K = 5$ and $\alpha = 0.05$. The black curve plots the distribution under H_0, which is $F(J, T - K - 1; \lambda = 0)$; while the red curve plots the distribution under H_1, which is $F(K, T - K - 1; \lambda > 0)$, where the value of λ is calculated using (3). Let $F_{c,\alpha}$ denote the α-level critical value from $F(J, T - K - 1; \lambda = 0)$. The blue vertical line is associated with the 0.05-level critical value $F_{c,0.05}$ of 2.22. The grey-shaded area under the black curve represents α, the probability of Type I error; while the area under the red curve below $F_{c,\alpha}$ represents the probability of Type II error, denoted as β. The red-shaded area represents the power of the test $\Pi \equiv 1 - \beta$. The p-value of the test for H_0 is defined as $p \equiv Pr(F > F_{c,\alpha} | \lambda = 0)$, where F denotes the observed F-statistic. Under the p-value criterion, H_0 is rejected at the α level of significance if $p < \alpha$; or alternatively, if $F > F_{c,\alpha}$.

2.2 *p-Value as a Measure of Evidence*

The p-value measures how unlikely the observed statistic is if H_0 were true. As the American Statistical Association (ASA) (Wasserstein and Lazar, 2016) points out, a p-value does not provide a good measure of evidence regarding a model or hypothesis. While the p-value can indicate how incompatible the data is with H_0, it does not measure the probability that H_0 is true, nor does it measure the importance of the variable being tested under H_0 (Wasserstein and Lazar, 2016). On this basis, the ASA makes clear that 'scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold'.

In fact, as Berger and Sellke (1987) show, actual evidence against H_0 can differ by an order of magnitude from the p-value. Johnstone and Lindley (1995) make a similar point, demonstrating that a p-value less than 0.05 may represent evidence that favours the null hypothesis, not against it, especially when the sample size is large. Revisiting the example used in Kim *et al.* (2018), suppose the observed F-statistic is 3 in Figure 1 when $T = 5000$. The p-value is $Pr(F > 3|H_0) = 0.01$ (from the black curve), leading to rejection of the null at 0.05 level. However, $Pr(F < 3|H_1) = 0.0001$ (from the red curve), much smaller than the p-value. This means that the F-test outcome is incompatible with H_0 with an odd of 0.01, while it is incompatible with H_1 with a substantially lower odd of 0.0001. That is, the p-value of 0.01 in fact indicates strong evidence for H_0, not against it, in comparison with the evidence for H_1. On this point, Startz (2014; p. 124) states that 'We should be concerned with the relative probabilities of the competing hypotheses'. The ratio of the two probabilities is called the Bayes factor (introduced later), a measure of evidence widely used in the Bayesian method.

2.3 *Zero Probability Paradox and Large-Sample Problem*

The statistical theory presented in Section 2.1 is based on the assumption of H_0 holding exactly true. However, in a real-world research, the probability of a point null hypothesis holds true exactly is zero, as shown by Rao and Lovric (2016) in the context of testing for the population mean. When H_0 is true, it is more realistic that it is practically true with an economically negligible deviation; but it is highly unlikely that it holds exactly and literally. This point has been made by many authors such as Hodges and Lehmann (1954), Leamer (1988) and De Long and Lang (1992). In particular, Startz (2014; p. 122) states that 'Economic hypotheses are usually best distinguished by some parameter being small or large, rather than some parameter being exactly zero versus non-zero'. As an example, let Y be a stock market return and X_1 is a factor with γ_1 denoting its coefficient. $H_0 : \gamma_1 = 0$ means that the response to stock return to a factor is 0 exactly without any deviation, which is unrealistic. Grossman and Stiglitz (1980) show that a perfectly efficient market is impossible because if prices fully reflect all available information, traders would not have any incentive to acquire costly information. Hence, in reality, the true value of γ_1 can be close to 0, say -0.001, even if H_0 is true.

Rao and Lovric (2016) call this problem 'zero probability paradox' of a point null hypothesis testing, stating that this is a root cause of the 'large n problem'. A consequence of this paradox is that, in practice, the test statistics are always generated from the distribution under H_1, not from that of H_0. That is, as demonstrated in the second panel of Figure 1, the p-value criterion can deliver rejection H_0 almost always with a sufficiently large sample, although it is practically true with an economically negligible effect ($R^2_{p1} = 0.01$). In this case, while H_0 is practically true, the probability of rejecting it (false positive) increases with sample size (see, also, Kim and Ji, 2015, section 5; Ohlson, 2018). As Spanos (2017) points out, this is a reflection of the consistency property of a statistical test. Rao and Lovric (2016) assert that there is no cure for this under the current point-null paradigm, suggesting that the researchers consider an alternative to the p-value criterion, especially when the sample size is large or massive.

In addition, the p-value criterion is unable to balance the dominant effect of the sample size over the model's explanatory power. It is clear from (2) and (3) that the F-statistic and its non-centrality increase with the sample size at the rate of T; or, equivalently, the p-value is decreasing with sample size. Since R^2 (and R^2_p) does not increase with sample size, even when they are tiny (or the model shows a negligible explanatory power), the F-test statistic can be inflated with a sufficiently large sample size to reject H_0, if the value of α is maintained at a conventional one such as 0.05. This point is again clear from the second panel of Figure 1. On this basis, Arrow (1960) and Leamer (1978) maintain that the level of significance should be adjusted as a decreasing function of sample size, a point advocated by a number of finance researchers such as Klein and Brown (1984), Neal (1987), Connolly (1989, 1991) and Kim and Ji (2015).

Another problem of large-sample analysis is that the researchers can lead to what is called the ‘fallacy of rejection’. It refers to the situation where the evidence against H_0 is misinterpreted as evidence for a particular H_1 (Spanos, 2017; section 4.2.3). In our example given in Section 2.1, where Y is stock return and X_1 is a factor, the fallacy means that rejection of $H_0 : \gamma_1 = 0$ is interpreted as direct evidence against market efficiency. The evidence for or against market efficiency also depends on the economic plausibility of X_1 as a factor (Harvey, 2017); and whether the true value of γ_1 is meaningfully different from 0 (McCloskey and Ziliak, 1996), among others. The fallacy often occurs when the sample size is large or massive, where H_0 can be rejected with a substantively tiny discrepancy due to extreme power of the test. If the researcher wishes to maintain a large or massive sample size, she should focus on whether the effect size estimate is economically meaningful or whether their X variable of interest is making a meaningful contribution to the variation of Y, indicated by the incremental signal-to-noise ratio η given in (5). In addition, visualization of the data using a range of graphical methods is strongly recommended.

3. A Statistical Toolbox

In making statistical decisions based on hypothesis testing, this paper proposes that the researchers use a statistical toolbox which contains a range of alternative instruments, given the shortcomings of the *p*-value criterion discussed above. In particular, Gigerenzer (2004; p. 604) states that

> We know but often forget that the problem of inductive inference has no single solution. There is no uniformly most powerful test, that is, no method that is best for every problem. Statistical theory has provided us with a toolbox with effective instruments, which require judgment about when it is right to use them. ... Judgment is part of the art of statistics.

In this section, a number of simple and effective instruments that an empirical researcher can have in her statistical toolbox are introduced. They include those based on the classical method, Bayesian method, equivalence testing and those as a compromise between the two, as well as simple graphical methods.

3.1 *Bayes Factor*

From the discussion given above, it is clear that the Bayesian inference is a useful alternative to the classical hypothesis testing based on the *p*-value criterion. It is based on the posterior odds ratio defined as

$$P_{10} \equiv \frac{Pr(H_1|D)}{Pr(H_0|D)} = \frac{Pr(D|H_1)Pr(H_1)}{Pr(D|H_0)Pr(H_0)} \tag{6}$$

where D denotes the data, $Pr(H_i)$ the prior probability of H_i, $Pr(D|H_i)$ the marginal distribution of data under H_i and $Pr(H_i|D)$ the posterior distribution for H_i. The evidence favours H_1 over H_0 if $P_{10} > 1$. Note that $B_{10} \equiv \frac{Pr(D|H_1)}{Pr(D|H_0)}$ is called the Bayes factor, which is equal to P_{10} when $Pr(H_0) = Pr(H_1)$. It is clear from its definition that the Bayes factor explicitly compares the compatibility of data with H_0 and H_1. Kass and Raftery (1995, p.777) classify that the evidence against H_0 is ‘not worth more than a bare mention’ if $2log(B_{10}) < 2$; ‘positive’ if $2 < 2log(B_{10}) < 6$; ‘strong’ if $6 < 2log(B_{10}) < 10$; and ‘very strong’ if $2log(B_{10}) > 10$, where log denotes the natural logarithm.

Zellner and Siow (1980) derive a Bayes factor for a regression F-test (or t-test) for model (1), based on a Cauchy distribution for the prior, which can be written as:

$$B_{10} = \frac{\Gamma[0.5(k_0+1)][1+(k_0/v_1)F]^{0.5(v_1-1)}}{\pi^{0.5}(0.5v_1)^{0.5k_0}} \tag{7}$$

where $\Gamma()$ is the Gamma function and $v_1 = T - k_0 - k_1 - 1$, while k_0 is the number of parameters restricted under H_0 and k_1 is the number of parameters unrestricted (i.e. $K = k_0 + k_1$). A Bayes factor can also be calculated from the R^2 value of a regression. Assuming a Cauchy prior of Zellner and Siow (1980) on the regression coefficients, Liang *et al.* (2008) derive

$$B_{10} = \int_0^{\infty} (1+g)^{0.5(T-K-1)}[1+g(1-R^2)]^{-0.5(T-1)}\theta(g)dg \tag{8}$$

where $\theta(g) = \frac{\sqrt{0.5T}}{\Gamma(0.5)} g^{-1.5}\exp(-T/2g)$ and g is the scale parameter for the regression coefficients which follows an inverse-gamma distribution. This Bayes factor is applicable to a test for the null hypothesis that all slope coefficients of a regression model are jointly equal to 0. Hence, it may be regarded as a Bayesian equivalent to the regression F-statistic for the joint significance of all explanatory variables of the model. According to Liang *et al.* (2008), the above integral can be carried out using a standard integration technique or using the Laplace approximation. Its computational resources are available from a package called ‘BayesFactor’ (Morey and Rouder, 2015) written in *R* (R Core Team, 2017). An attractive feature of the above Bayes factors is that they are simple functions of R^2 or the F-test statistic, allowing for a linkage between the classical and Bayesian methods of statistical inference.

3.2 *Probability that H_0 is True*

In expression (6), $Pr(D|H_0)$ represents the p-value, which denotes the probability of the sample (estimate) given that H_0 is true; or how incompatible it is with H_0. The probability that H_0 is true given the sample or after the researcher observes the data, denoted as $Pr(H_0|D)$ in (6), would be a more useful and informative indicator to empirical researchers in their statistical decision making. Applying the Bayes theorem to the linear regression model, Startz (2014) proposes the following easy-to-apply formulae for the (posterior) probability that H_0 is true, given the sample estimate of the parameter of interest:

$$P_{H_0} \equiv Pr(H_0|\hat{\gamma}) \approx \frac{\phi(t)}{\phi(t) + [c/\sigma_{\hat{\gamma}}]^{-1}} \tag{9}$$

where t denotes the usual t-statistic for $H_0 : \gamma = \gamma_0$ against $H_1 : \gamma \neq \gamma_0$, $\phi(\cdot)$ is the standard normal density, $\hat{\gamma}$ denotes the estimator for γ, $\sigma_{\hat{\gamma}}$ its standard error estimator and c is a constant chosen by the investigator as the width of an interval centred on γ_0. According to Startz (2014), the researcher can choose the interval as the *ex ante* region of interest under H_1; or she can adopt a simple rule $c = \sqrt{2\pi T\sigma_{\hat{\gamma}}}$ for $T > 60$ which is equivalent to using the *BIC* for model selection. As the second best practice to (9), Startz (2014) recommends using

$$P_{H_0} \approx \frac{\exp(0.5[-t^2 + \log(T)])}{1 + \exp(0.5[-t^2 + \log(T)])} \tag{10}$$

which is a large sample approximation to P_{H_0} based on the *BIC*. Startz (2014; section 9) shows that P_{H_0} has a desirable property of consistency: it converges to 1 under H_0, but to 0 under H_1, in large samples. Startz (2014) argues that P_{H_0} is a relevant measure for distinguishing between economic hypotheses, since it is formulated considering the distributions under H_1.

3.3 *Adaptive Level of Significance*

It has been argued that keeping the level of significance at a constant value is inappropriate (Arrow, 1960); and it should be set as a decreasing function of sample size (Leamer, 1978). From statistics, Perez and Perichhi (2014) propose a simple adaptive rule for the level of significance by reconciling the Bayesian and likelihood ratio principles. In the context of $H_0 : \gamma_1 = \cdots = \gamma_J = 0$, the rule can be written as:

$$\alpha^+ = \frac{[\chi^2_\alpha(J) + J\log(T)]^{(0.5J-1)}}{2^{(0.5J-1)}T^{0.5J}\Gamma(0.5J)}\exp[-0.5\chi^2_\alpha(J)] \tag{11}$$

which adjusts the level of significance as a decreasing function of sample size. Since this rule is derived in connection with the Bayes factor, the power of the test is implicitly taken into account.

3.4 *Optimal Level of Significance*

The level of significance can be chosen optimally by minimizing the expected loss from hypothesis testing. The proposal has been made by a number of authors including Arrow (1962), Leamer (1978) and Das (1994); see Piorier (1995) for a review. Recent studies that propose the use of the optimal level of significance include Mudge *et al.* (2012), Weakliem (2016) and Kim and Choi (2017, 2018). The trade-off between α and β is well known: a higher value of the former is associated with a lower value of the latter, and *vice versa*. Following Leamer (1978), from all possible combinations of α and β, the optimal value of α is determined by minimizing the expected loss from hypothesis testing, which can be written as:

$$EL(\alpha; P, k) = P\alpha + (1 - P)k\beta \tag{12}$$

where P is the (prior) probability that H_0 is true and $k \equiv L_2/L_1$ is the relative loss with L_i being the losses of Type i errors (i = I or II). If the researcher believes that H_0 is highly likely, then a high value of P may be specified. Similarly, if the researcher is much more cautious about Type I error than Type II, then a higher value is assigned to L_1 than to L_2, specifying a low value of k. When the researcher is neutral between H_0 and H_1, in terms of prior belief and relative loss, she can set $P = 0.5$ and $k = 1$, which means that the researcher believes that H_0 and H_1 are equally likely; and the losses of Type I and II errors are equal. Under these assumptions, the expected loss to be minimized is $\alpha + \beta$. The optimal level which minimizes the expected loss is denoted as (α^*, β^*).

Figure 2 presents the lines which plot all possible combinations of (α, β) from a regression t-test (one-tailed) for a range of sample size, which Leamer (1978) calls the line of enlightened judgement. The line shifts towards the origin with a larger sample size and a higher power. The points of intersection between the lines of enlightened judgement and 45 degree line are the points (α^*, β^*) where the expected loss $\alpha + \beta$ is minimized. As it is clear, there is no reason to believe that the conventional levels of significance such as 0.05 and 0.01 are optimal. In fact, by adopting these conventional levels, a researcher is faced with a higher expected loss from her decisions.

In practice, it is often the case that a researcher may not be impartial between the null and alternative hypotheses, where $P \neq 0.5$. For example, economic theory may suggest that a particular point of the parameter space is highly likely. Based on this, the researcher can specify a particular value of P. Similarly, a researcher may specify a particular value of relative loss k, depending on her attitude towards Type I and II errors: see, for economic applications with the general values of P and k, Kim and Choi (2018).

In calculating the values of β given the values of α as illustrated in Figure 2, the point of the parameter space under H_1 should be determined. If this value is too close to the value under H_0, the power will be too low; but if this value is too far away from it, the power can be too high or even equal to 1. This point should represent an economically meaningful deviation from H_0, which is called the 'minimum oomph' (Ziliak and McCloskey, 2008) or 'critical effect size' (Mudge *et al.*, 2012). The choice of

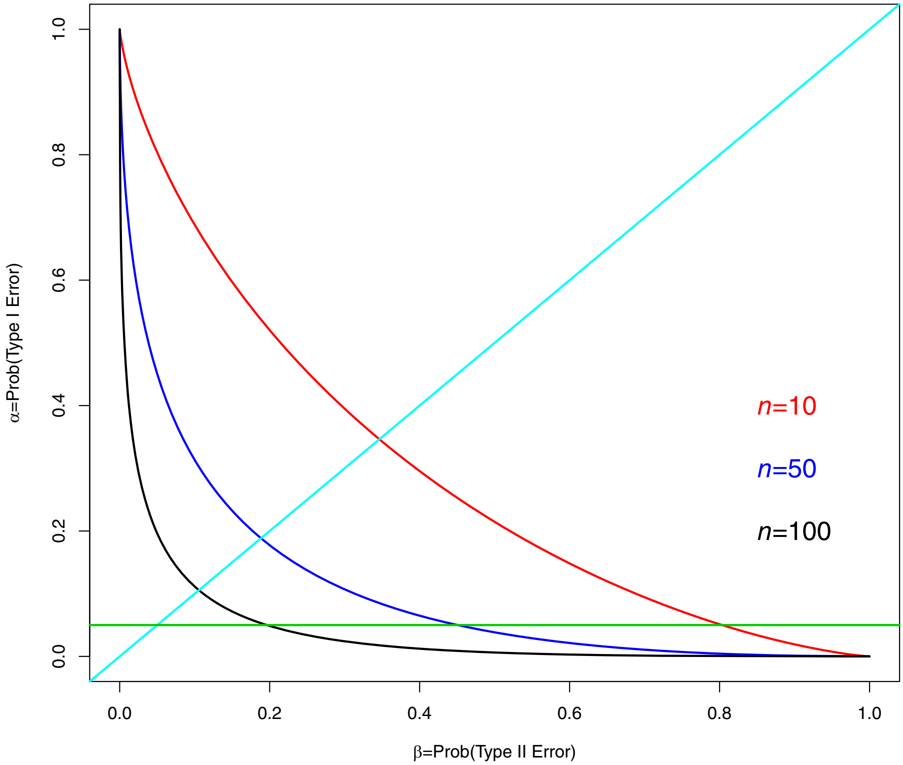

Figure 2. Examples of the Line of Enlightened Judgement. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The 45-degree line corresponds to the points, where $\alpha + \beta$ is minimized, and the horizontal one to $\alpha = 0.05$.

this value may require a careful investigation of the effect size. When such a choice is not clear or arguable, Kim and Choi (2018) recommend that researchers try a number of alternative values in the neighbourhood of sample estimate $\hat{\gamma}_j$. For a joint test such as $H_0 : \gamma_1 = \cdots = \gamma_J = 0$, the power can be calculated based on the non-centrality parameter λ. That is, we may use the sample estimate non-centrality parameter $\hat{\lambda} = T\frac{(R_1^2 - R_0^2)}{1-R_1^2}$ or find the minimum value of λ that represents a meaningful deviation from 0, denoted as,

$$\lambda^* = T\frac{R_{*1}^2 - R_{*0}^2}{1 - R_{*1}^2} \tag{13}$$

where R_{*j}^2 denotes the desired coefficient of determination under H_j. Then, the value of β can be calculated using the value of λ^* or in the neighbourhood of it. Further technical details, such as the weighted optimal level of significance, can be found in Kim and Choi (2018). Kim (2017a) provides computational resources (package 'OptSig') written in *R* (R Core Team, 2017), for the optimal level of significance for a range of statistical tests including the *t*- and *F*-tests.

3.5 *Equal-Probability Test and Sample Size Selection*

From the above discussions, it is clear that a sensible research design should involve a balance between the probabilities of Type I and II errors. Arrow (1960) and Naik (1969) propose the 'equal-probability test', where the level of significance (α) is set equal to Type II error probability (β). To formalize the test in the context of the F-test, the equal-probability test satisfies $\alpha = \beta$. That is,

$$Pr(F(J, T - K - 1; \lambda = 0) > F_{c,\alpha}) = Pr(F(J, T - K - 1; \lambda = \lambda^*) < F_{c,\alpha}) \tag{14}$$

An equal probability test is illustrated in the first panel of Figure 1, where $\alpha = \beta = 0.05$ with $\Pi = 0.95$. In this case, the test shows a perfect balance between the two error probabilities, as well as a sufficiently high power. Given the chosen values of $\alpha(= \beta)$ and λ^*, the required sample size (denoted as T^*) for the equal-probability test can be determined. As an example, in Figure 1, suppose one wishes to conduct the F-test with $\alpha = \beta = 0.005$ and $\Pi = 0.995$. Suppose further that, for the model to be economically significant, it is required that $R^2_{*1} = 0.1$ ($R^2_{*0} = 0$). Then, it can be determined from (14) that $T^* = 360$ (approximately). Obviously, the attraction of this test is that the researcher has a full awareness and control of the Type I and II error probabilities (and the power of the test). In fact, this is a part of the original proposal of the significance test of Neyman–Pearson (see Gigerenzer, 2004). Note that the equal-probability test can also be conducted in the context of the regression t-test in the same way.

3.6 *Cautionary Analyses for Large-Sample Bias*

Using large or massive sample is widespread in empirical research in business, especially in accounting and finance (see Kim and Ji, 2015; and Kim *et al.*, 2018). While large data sets provide powerful analytical tools and new research opportunities, there are growing concerns that big data is often subject to a range of data issues such as sampling error, measurement error, selection bias, missing information and aggregation error: see, for example, Kaplan *et al.* (2014). A number of authors warn that these issues have the potential to provide misleading research outcomes and decisions; see, for example, Boyd and Crawford (2011), Wigan and Clarke (2013), Harford (2014), Hand (2016, 2018) and Fiebig (2017).[1]

In particular, Harford (2014) and Hand (2016) make a point that gathering more and more data in an effort to achieve 'T = ALL' is impossible, with potentially undesirable consequences. This also violates the notion of random sampling underlying the statistical theory of inference, as Hand (2018; p. 567) points out. Fiebig (2017) argues that big data is not synonymous with better or more informative data, pointing out that it is often subject to missing observations and measurement errors. In addition, a large data set may include extensive cross-sectional and time-series variations and noises, which cannot be fully captured by a single regression model using the whole sample. On this point, Hand (2016; p. 631) cautions that big (data) does not necessarily mean accurate or comprehensive, emphasizing the importance of the underlying theory and analysis of small data sets. Kaplan *et al.* (2014) note that, with a large data set, 'investigators will surely have less intimate knowledge about the texture and often the quality of the many elements in their datasets'.

As Box (1976) emphatically asserts, 'Essentially, all models are wrong, but some are useful'. This means that a model is a simplification or approximation of the reality, and all model estimation results are subject to specification error bias. In the context of specification testing, with a large enough sample size, all statistical models will be judged to be mis-specified (Spanos, 2017). Under model mis-specification, it is well known that the parameter estimator is biased and inconsistent. As Kaplan *et al.* (2014) point out, when the sample size is small, the estimation bias may not show up strongly; but as the sample size grows, the estimation bias can be magnified. Even when the model is useful initially, as the sample size grows to include a higher number of cross-section units or time dimension, it is possible that the data include additional cross-sectional variations, structural changes and noises, which cannot be captured fully by a single econometric equation. That is, the degree of model specification bias can increase as the researcher includes additional data points.

Turning to the issue of statistical inference, none of the inferential methods discussed so far is fully capable of providing an effective guard against the large-sample problem (see, for example, Weakliem, 2016). As an example, consider a simple regression between Y and X_1 with $H_0 : \gamma_1 = 0$. Suppose the true value of γ_1 is small with $R^2_{p1} = 0.01$ (note that $R^2_{p0} = 0$). According to (2), the F-statistic (or t) is greater than its 5% critical value as long as $T \geq 382$. The Bayes factors in (7) and (8) also provide 'positive' evidence against H_0 so long as $T \geq 1200$. Hence, these inferential statistics will eventually reject H_0 if a researcher increases the sample size, even though the model can explain a negligible variation of Y. That is, as Ohlson (2018) points out, the likelihood of a false positive increases with sample size (and power). As discussed in Section 2, this occurs as a result of the consistency property, with their statistical power being practically 1 under these sample sizes, as also illustrated in the second panel of Figure 1.

It should also be noted that these inferential methods are not intended for large or massive samples. Keuzenkamp and Magnus (1995; p.17) point out that Fisher's theory of significance testing is intended for small samples, stating that 'Fisher does not discuss what the appropriate significance levels are for large samples', a point also made by Lindley (2014; section 14.4). Meanwhile, the values of α^* (optimal significance level discussed in Section 3.4) and α^+ (adaptive significance level given in (11)) quickly approach zero as the sample size increases, providing a useful guard against the problem. However, a research design in which the level of significance is practically set at 0 is not statistically sound, since it does not allow for Type I error to occur, while the probability of Type II error also approaches 0 with increasing sample size.

The discussions in this and previous sub-sections suggest that empirical researchers with large samples be aware of the potential large-sample biases that can possibly mislead their research outcomes. Cautionary basic data analyses with purposeful attention to these large-sample biases and appropriate sensitivity and graphical analyses should be a part of their statistical toolbox (Kaplan *et al.*, 2014; Hand, 2016; Rao and Lovric, 2016; Ohlson, 2018). It is also desirable that a smaller sample size should be adopted (see Hand, 2016; Ohlson, 2018). This does not mean that we should throw data away, but it suggests that the researchers conduct a range of meaningful sub-sample analyses using a smaller sample size (sample size selection as discussed in the previous sub-section). By doing this, a researcher is able to conduct the inference for a range of sub-samples or random samples, with a full awareness of α and β (and power) values, implementing a range of alternative tools in the toolbox.

3.7 *Equivalence Testing*

In light of the zero probability paradox for a sharp null hypothesis and the large-sample problems associated, Rao and Lovric (2016) argue that a new paradigm in statistical inference should be in place for statistical research in the 21st century, given that large or massive data sets are becoming more and more available. As a natural alternative to the conventional point-null hypothesis testing, Rao and Lovric (2016) propose a test for equivalence (or a negligible null hypothesis), which was an original proposal of Hodges and Lehmann (1954). This is consistent with the points made by a number of authors that an economic hypothesis should be formulated as an interval, not as a point; see Leamer (1988), De Long and Lang (1992) and Startz (2014). With equivalence testing, the null and alternative hypotheses are formulated as intervals. That is, following Lauzon and Caffo (2009), to test for the significance of the parameter γ, the null and alternative hypotheses written as:

$$H_0 : (\gamma_l \geq \gamma) \cup (\gamma_u \leq \gamma); H_1 : \gamma_l < \gamma < \gamma_u \tag{15}$$

where $\gamma_l < 0$ and $\gamma_u > 0$ are, respectively, lower and upper tolerance limits of economic significance of γ. These values should be pre-determined by the researcher based on economic analysis and reasoning, which are often referred to as 'minimum oomph' (Ziliak and McCloskey, 2008) or 'critical effect size' (Mudge *et al.*, 2012). H_0 indicates the inequivalence between γ and zero, where the value of γ is of economic significance; while H_1 indicates the equivalence, where the value of γ is economically insignificant. The null hypothesis of inequivalence is rejected at the α level of significance if the $(100 - 2\alpha)$ level confidence interval lies entirely within the tolerance limits $[\gamma_l, \gamma_u]$ (see Lauzon and Caffo, 2009). The test for equivalence is highly effective as a guard against the large-sample problems discussed above, since the confidence interval shrinks as the sample size increases, while the tolerance limits do not as they are determined based on economic analysis or substantive importance of the relationship being tested.

3.8 *Further Remarks*

The instruments proposed above may show different characteristics, given the circumstances and context of research. As such, it would be informative to the readers if a guidance is provided comparing the alternative instruments in the proposed toolbox. In this sub-section, I provide recommendations as to which instrument would be more useful under different circumstances.

When the sample size is small or the power is low, the optimal level of significance (α^*) should be preferred to a conventional level of significance. When the sample size is moderately large, the instruments based on the Bayesian method (Bayes factor; posterior probability, P_{H_0}; and adaptive level of significance, α^+) should be useful alternatives to the *p*-value criterion.

When the sample size is massive, all of these inferential methods will be subject to the large-sample problem, as discussed in Section 3.3. For example, the values of α^+ and α^* would be practically zero with statistical power of 1 in this case, which does not represent sound research design since any negligible deviation from H_0 will be rejected. In this case, attention should be paid to the economic plausibility of the hypothesis (Harvey, 2017), careful analysis of effect size estimates (Ziliak and McCloskey, 2008) and the goodness-of-fit of the model or variable of interest (measured by η given in (5)). On this point, equivalence testing should be useful under a massive sample size, since its decision rule is based on the confidence interval and economic judgement on the effect size. In addition, it is highly recommended that the researchers employ a range of exploratory data analysis to gain visual impressions of the data. The researchers should also consider more extensive sub-sample analysis by selecting a reasonable sample size as discussed in Section 3.5. In this case, the researcher can conduct the test with a full awareness of the probabilities of Type I and II errors, by employing the optimal level of significance or equal-probability test.

It should be noted that the instruments in the toolbox proposed in this paper are by no means exclusive or exhaustive: the researchers can choose their own instruments depending on the problems at hand and the context of their research. For example, other candidates in the toolbox include those that can address model uncertainty, such as model averaging (see, for example, Moral-Benito, 2015), extreme bounds analysis (Sala-i-Martin, 1997); or those based on predictive inference widely used in big data analysis (see, for example, Lantz; 2015).

4. Applications

In this section, the statistical toolbox proposed in the previous section is applied to three notable studies in empirical finance. The first is the study on the stock return and the U.S. political cycle (Novy-Marx, 2014; Santa-Clara and Valkanov, 2003); the second is the effect of a major sporting events on stock market (Kaplanski and Levy, 2010); and the last is the relationship between firm-level investment and idiosyncratic risk (Panousi and Papanikolaou, 2012). These studies are selected on the basis of their importance in the literature, replicability and data availability.

4.1 *Stock Return and U.S. Political Cycle*

Novy-Marx (2014) reports the empirical results that a range of exotic factors (such as the U.S. president's party, weather, global warming, the El Nino phenomenon, sunspots and the conjunction of the planets) have predictive power for stock return. For the case of the U.S. presidency, Novy-Marx (2014) adopts the predictive regression of the form

$$Y_t = \gamma_0 + \gamma_1 X_{t-1} + e_t$$

where Y is the excess return (in percentage) and X is the dummy variable for the party of the U.S. president (1 for the Democratic president, 0 for the Republican). According to Novy-Marx (2014), the lagged specification is chosen 'following the literature'; while Santa-Clara and Valkanov (2003) state that it is chosen to emphasize that the political variables are known at the start of a return period. Note that Santa-Clara and Valkanov (2003) conduct a more comprehensive analysis on this issue, calling their result the 'Presidential puzzle'. They also adopt the robust standard error estimators based on asymptotic method and bootstrapping, but report that their inferential results do not change. Using the monthly U.S. data from 1961 to 2012 ($T = 642$), Novy-Marx (2014) reports that $\widehat{\gamma}_1 = 0.75$ with the t-statistic of 2.08, concluding that the party of the presidency has an economically significant effect on stock return.

4.1.1 *Replication and Effect Size*

Using the monthly excess return available from French's data archive over the same period, I replicate Novy-Marx's (2014) result almost exactly as follows:

$$\widehat{Y}_t = 0.12 + 0.76X_{t-1}$$

with the t-statistic for $\widehat{\gamma}_1$ being 2.11 (p-value=0.035) and $R^2 = 0.007$. The intercept estimate of 0.12 is statistically insignificant at any reasonable level of significance with the p-value of 0.64. The result may represent a conflict between estimation and statistical inference, since a statistically significant predictor (at the 5% level) can explain only 0.7% of the total variation of the excess stock return. This low value of R^2 may also be a sign of model mis-specification, which makes the validity of statistical inference questionable.

While the expected monthly excess return is found to be 0.76% higher during the Democratic presidency, whether this effect size is economically large enough may be arguable, given that the model's explanatory power is tiny with $\hat{\eta} = 0.007$ (signal-to-noise ratio). Santa-Clara and Valkanov (2003) state that the expected return during the Democratic presidency is 9% per year ($\simeq 0.76 \times 12$), but this assessment does not take account of the underlying uncertainty of stock return. As Figure 3 shows, the expected monthly return of 0.76% under the Democratic presidency may not be economically significant, considering the high volatility of stock return, high downside risk and possible trading costs. The Box-and-Whisker plots presented in Figure A1 of the Appendix show that the return distributions have nearly identical values of the quartiles during the two periods, with the median values statistically indistinguishable. It appears that there are more extreme return values during the time of the Republican presidency, which may have distorted the regression results.

4.1.2 *Alternatives to the p-Value Criterion*

The Bayes factor does not favour H_1, with the value of $2log(B_{10})$ calculated from the Zellner–Siow formula given in (7) being -1.227, indicating that the association is 'not worth more than a bare mention'. According to the Liang *et al.* (2008) formula given in (8), the value of $2log(B_{10})$ is -0.280, delivering the same inferential outcome. The values of P_{H_0} based on (9) and (10) are, respectively, 0.82 and 0.98,

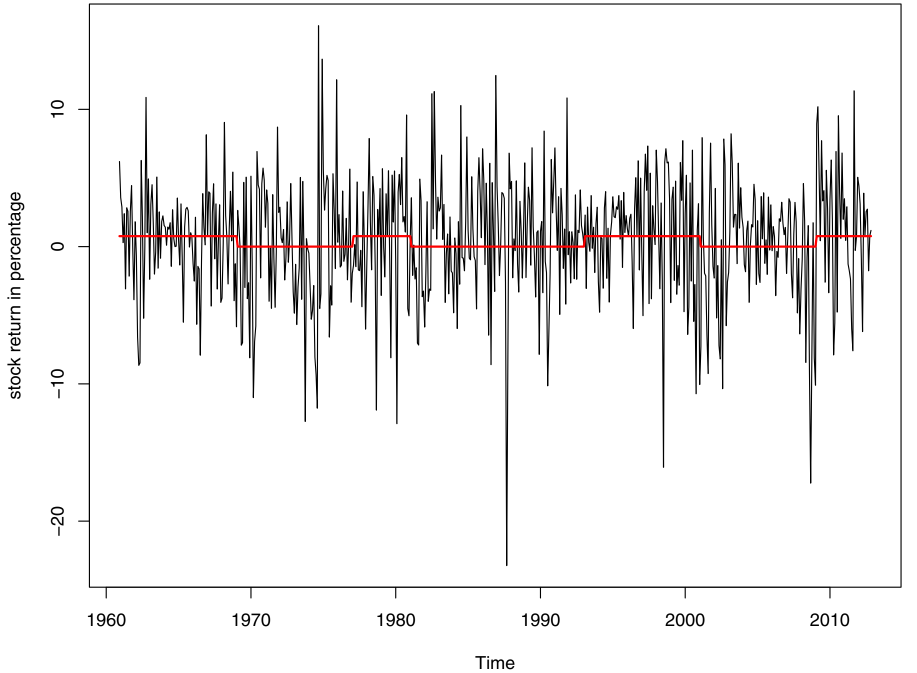

Figure 3. Expected Return during Democratic Presidency. [Colour figure can be viewed at wileyonlinelibrary.com]
Note: The red line indicates 0 during the Republican term and 0.76 during the Democratic term.

indicating that H_0 is highly likely given the sample estimate of γ_1. The adaptive level of significance α^+ using the Perez–Pericchi (2014) formula given in (11) is 0.00146, well below the p-value of 0.035, in favour of no rejection of H_0, consistent with the Bayesian alternatives. Hence, these alternatives to the p-value criterion indicate that the political cycle does not have a statistically significant effect on stock return.

4.1.3 *Further Analyses*

The value of $2log(B_{10})$ calculated from the Liang *et al.* (2008) formula is 6.68, when $R^2 = 0.03$ and $T = 642$, corresponding to 'positive' evidence against H_0. This means that a Bayesian alternative would support H_1 when the value of R^2 is as large as 0.03. With this value of R^2, I design an equal probability test at the optimal level of significance for $T = 642$, with $\alpha^* = \beta^* = 0.02$ (approximately) and $\Pi = 0.98$. This power is substantially higher than that of 0.55 ($\beta = 0.45$) evaluated under $R^2 = 0.007$, $T = 642$ and $\alpha = 0.05$. Figure 4 presents the lines of the enlightened judgement for each case indicated with the points where the expected loss $\alpha + \beta$ is minimized. Under the equal-probability test, the two error probabilities are balanced with a substantially higher power. Moreover, the expected loss ($\alpha + \beta$) is 0.04, about 10 times smaller than the expected loss of 0.50 under $\alpha = 0.05$ and $R^2 = 0.007$. This means that a researcher who wishes to minimize the expected loss from Type I and II errors would surely prefer the research design under the equal-probability test. Since $\alpha^* = 0.02$ is less than the p-value of 0.035, we cannot reject H_0 under the equal-probability test, in accordance with the outcomes based on the Bayes factors.

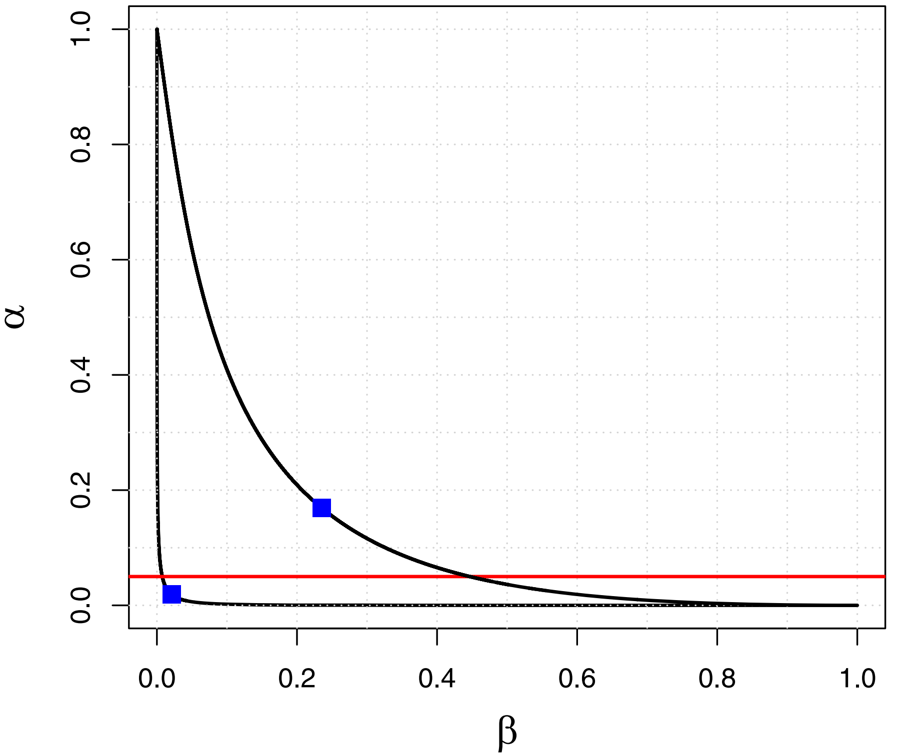

Figure 4. An Equal Probability Test for Presidential Cycle Regression. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The inner black curve is the line of enlightened judgement associated with $R_*^2 = 0.03$ with the blue dot indicating $\alpha^* = \beta^* = 0.02$ (approximately). The outer black curve is the same associated with $R_*^2 = 0.007$ with the blue dot indicating $\alpha^* = 0.17$ and $\beta^* = 0.24$.

Figure 5 plots the values of the optimal level of significance α^* against the values of R_{*1}^2, with the horizontal line indicating the p-value of 0.035. At $R_{*1}^2 = 0.007$, $\alpha^* = 0.167$, indicating the rejection of H_0 since α^* is greater than the p-value of 0.035. At $R_{*1}^2 = 0.03$, $\alpha^* = 0.02$, which leads to the acceptance of H_0, similarly to the equal probability test discussed above. The turning point occurs at the intersection of the two lines, where $R_{*1}^2 = 0.023$. This means that we can reject H_0 of no association only when the political cycle explains less than 2.3% of the total variation of the stock return. The value of γ_1 associated with this value of R_{*1}^2 is 1.33, calculated using (4), which is much larger than the sample estimate of 0.76. One may argue that these required values under H_1 ($R_{*1}^2 = 0.023$ and $\gamma_1 = 1.33$) are too high. However, it is doubtful that a factor that can explain less than 2.3% of the total variation of stock return can be regarded as economically important, given the descriptive properties of stock return observed in Figures 3 and A1. These values under H_1 may be subject to further economic analysis and judgement of the researcher, and this should be the target of a rigorous and thorough economic analysis (McCloskey and Ziliak, 1996).

In light of the above result, I conduct a test for inequivalence with $H_0 : |\gamma_1| \geq 1.3; H_1 : |\gamma_1| < 1.3$, on the basis that an economically significant factor should explain at least 2% of the total variation of the

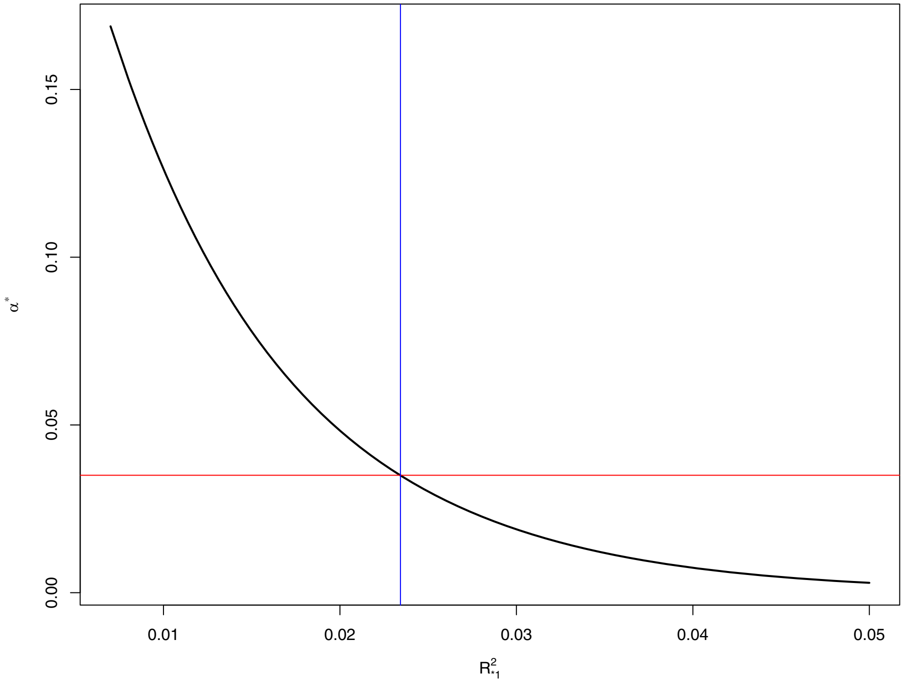

Figure 5. Optimal Level of Significance as a Function of R^2_{*1}. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The black curve indicates the optimal level of significance; the red curve indicates the horizontal line p-value $= 0.035$; and the blue curve indicates the vertical line with its intersection point of 0.023. Note that $R^2_{*0} = 0$.

stock return. The 80% confidence interval for γ_1 is found to be [0.30, 1.22], which lies entirely within the tolerance limit, rejecting H_0 of inequivalence at the 10% level of significance, which provides evidence that the estimated value of γ_1 is economically insignificant.

I demonstrate in this section that the statistically significant effect of the political cycle on stock return may be puzzling under the p-value criterion alone, but other more sensible alternatives in the toolbox point to no compelling evidence for its statistical and economic significance.

4.2 *Effect of a Major Sporting Event on the Stock Market*

It has been documented that major sports events have a systematic effect on stock (index) return with exploitable abnormal profits: see, among others, Edmans *et al.* (2007) and Kaplanski and Levy (2010). These findings are a part of the extensive literature, which claims that investors' moods (which are not related with the market fundamentals) have a significant effect on stock markets. To test for the FIFA World Cup effect on stock return, Kaplanski and Levy (2010) estimate the following equation:

$$R_t = \gamma_0 + \sum_{i=1}^{2} \gamma_{1i} R_{t-i} + \sum_{i=1}^{4} \gamma_{2i} D_{it} + \gamma_3 H_t + \gamma_4 T_t + \gamma_5 P_t + \gamma_6 E_t + \sum_{i=1}^{2} \gamma_{7i} J_{it} + \epsilon_t \tag{16}$$

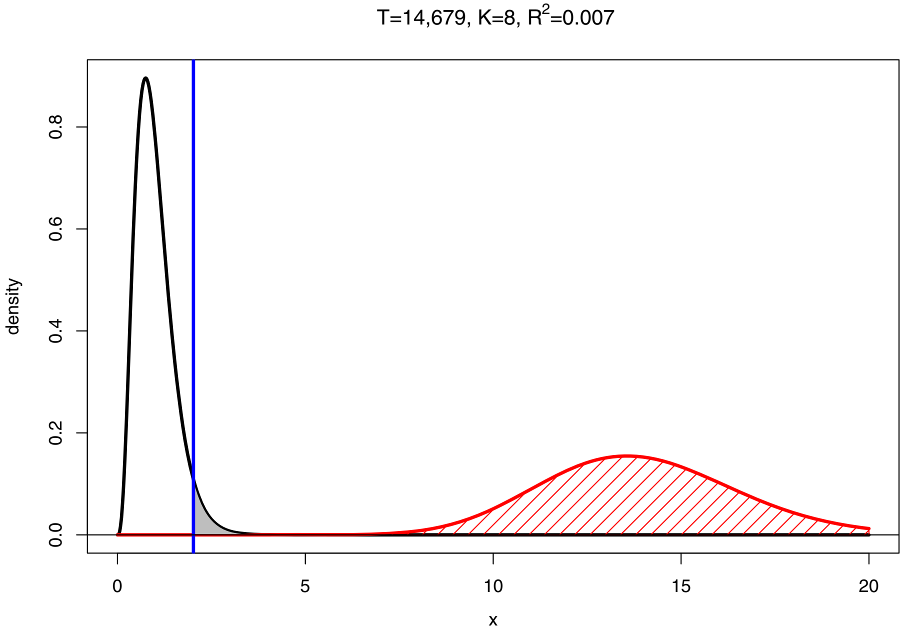

Figure 6. Distributions of F-Statistic under H_0 and H_1. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The black curve indicates the density of the F-test statistic under H_0; the red curve indicates the density under H_1. The blue vertical line is associated with the critical value at the 0.05 level of significance. The grey area represents the probability of Type I error (0.05) and the red shaded area represents the power of the test.

where R_t is the daily stock market return, D_{it} are the dummy variables for the day of the week (Monday–Thursday), H_t is a dummy variable for days after a non-weekend holiday, T_t is a dummy variable for the first 5 days of the taxation year, P_t is a dummy variable for the annual event period (June and July), E_t is the dummy variable for the event days and J_{it} are the dummy variables for the 10 days with the highest ($i = 1$) and lowest ($i = 2$) returns during the studied period. The model (16) follows those of prior studies such as Edmans *et al.* (2007) and Kamstra *et al.* (2003). For the dummy variable (E_t), Kaplanski and Levy (2010) use event effect days (EED) and event period effect days (EPED): the former being the game days that are trading days and subsequent trading days, while the latter being the all days of the FIFA World Cup (from the first game to the first day after the final game) plus two subsequent trading days. Kaplanski and Levy (2010) consider a few variations of (16) including an adjustment of the error term using GARCH model, but the results are found to be qualitatively similar to those of their base model. In this paper, I follow their based model (BM) setting $\gamma_{11} = \gamma_{12} = \gamma_{71} = \gamma_{72} = 0$ and $E_t = EPED$, due to its simplicity.

Using the daily returns from the NYSE composite index (CRSP) from 1950 to 2007 (14,679 observations), Kaplanski and Levy (2010; table 2) report a number of regression results based on (16), which indicate a statistically significant effect of FIFA World Cup games on stock returns. The estimated coefficients of γ_6 are around -0.002, and they are statistically significant with the t-statistic less than -3 in all cases, while the R^2 values are less than 0.1 in most cases. Figure 6 illustrates the distributions of the F-statistic in this case, which clearly shows that the test has the power of 1. Since the null hypothesis is always violated in practice, as discussed in Section 2, the F-statistic is generated from the red curve, not from the black one. In effect, this means that, if the researcher maintains the level of significance at a conventional value such as 0.05, H_0 is almost always rejected. The probability of Type I error is infinitely larger than that of Type II error which is zero in this case, which means that the research design adopted by Kaplanski and Levy (2010) is statistically flawed.

4.2.1 *Replication and Effect Size*

In this paper, the BM of (16) is estimated ($K = 8$) using the same return data updated to June 2016 ($T = 16,819$, value-weighted). It is found the FIFA World cup event coefficient estimate is -0.0015 with the t-statistic of -2.86 (two-tailed p-value of 0.0042), similar to those reported in Kaplanski and Levy (2010). The event coefficient is small indicating that the stock return is expected to be lower by -0.15% during the event days. While the value is statistically significant according to the p-value criterion, the value of R^2 is 0.005, which is tiny. The incremental contribution of the event dummy variable measured by $R_1^2 - R_0^2$ is 0.000485, indicating a negligible contribution of the FIFA World Cup to stock return variation. The Box-and-Whisker plots given in Figure A2 in the Appendix show that the return distributions share the nearly identical values of the median and other quartiles during the event and non-event periods. It appears that a substantially larger number of extreme returns are observed during the non-event period, which may have distorted the regression results.

4.2.2 *Alternatives to the p-Value Criterion*

The value of $log(B_{10})$ given in (3) for $H_0 : \gamma_6 = 0$ is -1.002, well below the threshold of 2 for being 'not worth more than a bare mention'. Both P_{H_0} values based on (9) and (10) are 0.99, indicating that H_0 is much more likely than H_1. The adaptive level of significance α^+ for $H_0 : \gamma_6 = 0$ is 0.0002, providing the one-tailed critical value of -3.48, leading to the acceptance of H_0. The optimal level significance α^* is also found to be practically 0. These results indicate that the relationship is not statistically significant if more sensible and stringent alternatives to the p-value criterion in the toolbox are adopted. For the purpose of equivalence testing, the 80% confidence interval for γ_6 is found to be $[-0.0022, -0.0008]$. This means that $H_0 : |\gamma_6| \geq 0.01$ is rejected at 10% level of significance in favour of $H_1 : |\gamma_6| < 0.01$. The tolerance limit 0.01 is chosen on the basis of an economic judgement that the stock return should be lower or higher by at least 1% during the FIFA World Cup period, for it to be economically significant event for stock market.

4.2.3 *Further Analyses*

In order to balance Type I and II error probabilities, I design an equal-probability test at the optimal level of significance. This means that the sample size should be reduced to an appropriate level. Figures 7 and 8 present such a case where $\alpha = \beta = 0.04$ ($\Pi = 0.96$) when $T^* = 2500$ for both t-test and F-test for the regression model (16). The F-test is to test for H_0 that all slope coefficient is jointly 0, with $R_{*1}^2 = 0.01$ and $R_{*0}^2 = 0$; and the t-test is for $H_0 : \gamma_6 = 0$, with $R_{*1}^2 = 0.01$ and $R_{*0}^2 = 0.005$. These values require that the model explains at least 1% of the total variation of stock return to be economically significant; and the event variable *EPED* should make a marginal contribution of at least 0.5% to the total variation of stock return.

Based on this design, I estimate the BM of (16) using the rolling sub-sample window of length 10 years (around 2500 observations including two FIFA World Cups). As reported in Figure 9, the estimated coefficients are in most cases negative, with the median of -0.0013 and the inter-quartile range of $(-0.0026, -0.0005)$. The R^2 values (not reported) are small over time with the median value of 0.0485

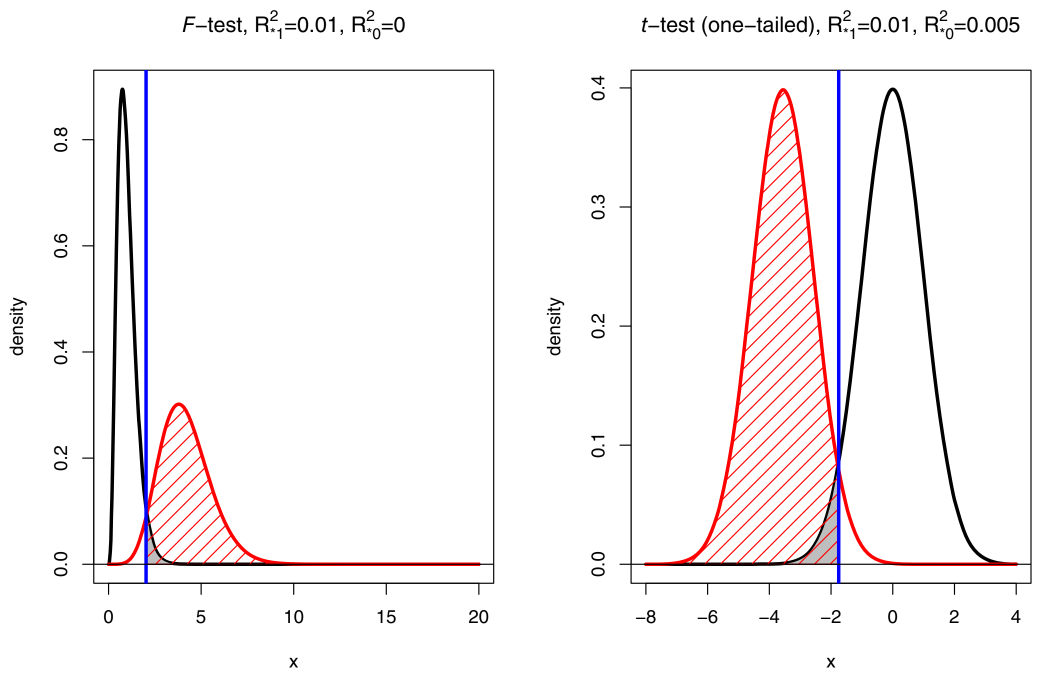

Figure 7. Graphical Representations of Equal-Probability Test. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The black curve indicates the density of the test statistic under H_0; the red curve indicates the density under H_1. The blue vertical line is associated with the critical value at the optimal level of significance of 0.04. The grey area represents the probability of Type I error (0.04); the red shaded area represents the power of the test of 0.96; and the white area under the red curve represents the probability of Type II error of 0.04 (T=2500, K=8).

and the inter-quartile range of (0.0274, 0.0532). According to the *t*-statistic, the event variable appears to be statistically significant at the 4% level, only from late 1960s to the mid-1980s. If the adaptive level of significance is adopted, the event variable is statistically significant only from the mid-1970s to the mid-1980s. The values of the $log(B_{10})$ are above 2 (threshold for a 'positive' effect) only from the mid-1970s to the mid-1980s, while they are all less than 6 (threshold for a 'strong' effect) for the entire period. Hence, the results indicate that the statistical significance of the FIFA World cup effect may hold only for about 10 years around 1980.

The estimated values of the EPED coefficient γ_6 indicate that the return is lower on average by around -0.1% per month during the FIFA World Cup. The maximum effect size from Figure 9 is -0.5% per month during from the mid-1970s to the mid-1980s. Again, given the volatility of daily stock returns, it is doubtful that these effect size estimates are economically significant. Although we observe some evidence that the FIFA World Cup may have statistically significance effect from the mid-1970s to the mid-1980s, the soccer did not gain popularity in the United States; and the United States did not participate in the World Cup finals during that period. Other factors such as the extent of foreign investment in the U.S. market (Kaplanski and Levy, 2010) and the media coverage of the event may have been much shallower and slower in the 1970s and 1980s compared to recent years. It is also possible that this period of statistical

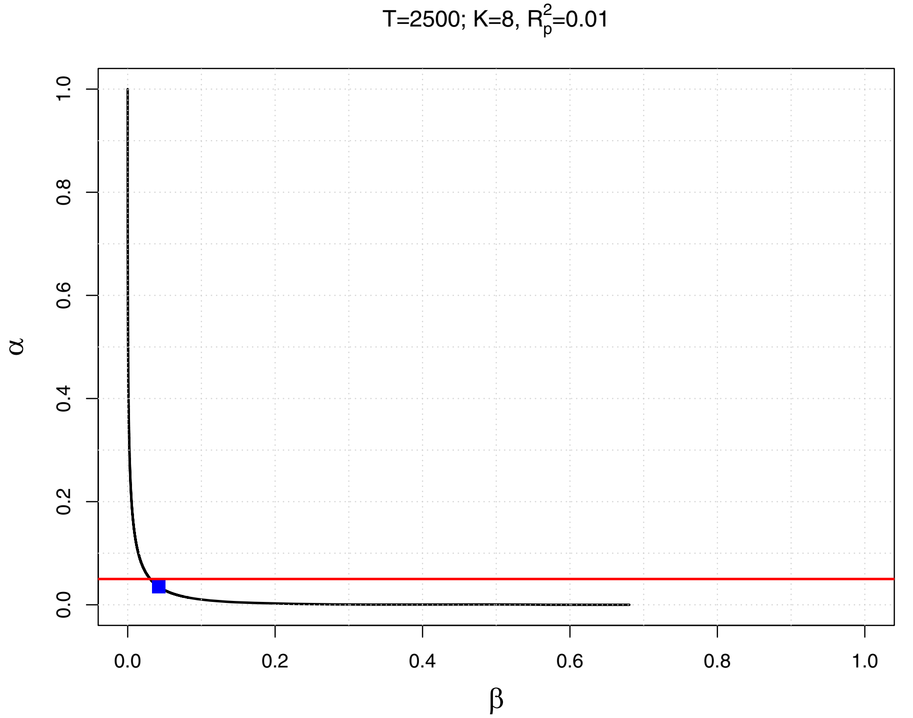

Figure 8. Line of the Enlightened Judgement and the Optimal Level of Significance. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The black line represents the line of the enlightened judgement, which plots all combinations of Type I-error probability and Type II-error probability for the main model of Kaplanski and Levi (2010). The blue dot is the point where the expected loss is minimized (0.04, 0.04), with the power of 0.96. The red line indicates the point of the conventional level of significance 0.05.

significance around 1980 has occurred due to data issues, such as the outliers around the oil shocks in the mid-1970s.

Overall, the results in this section strongly suggest that the existence of the economically and statistically significant FIFA World Cup effect on stock return is not quite convincing, when a range of alternatives in the toolbox are applied to the problem. Only the *p*-value criterion points to the statistical significant effect of the FIFA World Cup.

4.3 *Investment and Idiosyncratic Risk*

Panousi and Papanikolaou (2012) examine how the firm-level investment decision is related with the idiosyncratic risk using the following fixed-effects model of the form

$$\frac{I_{it}}{K_{it-1}} = \gamma_0 + \gamma_1 log(\sigma_{it-1}) + \gamma_2 Z_{it-1} + \eta_i + g_t + u_{it} \tag{17}$$

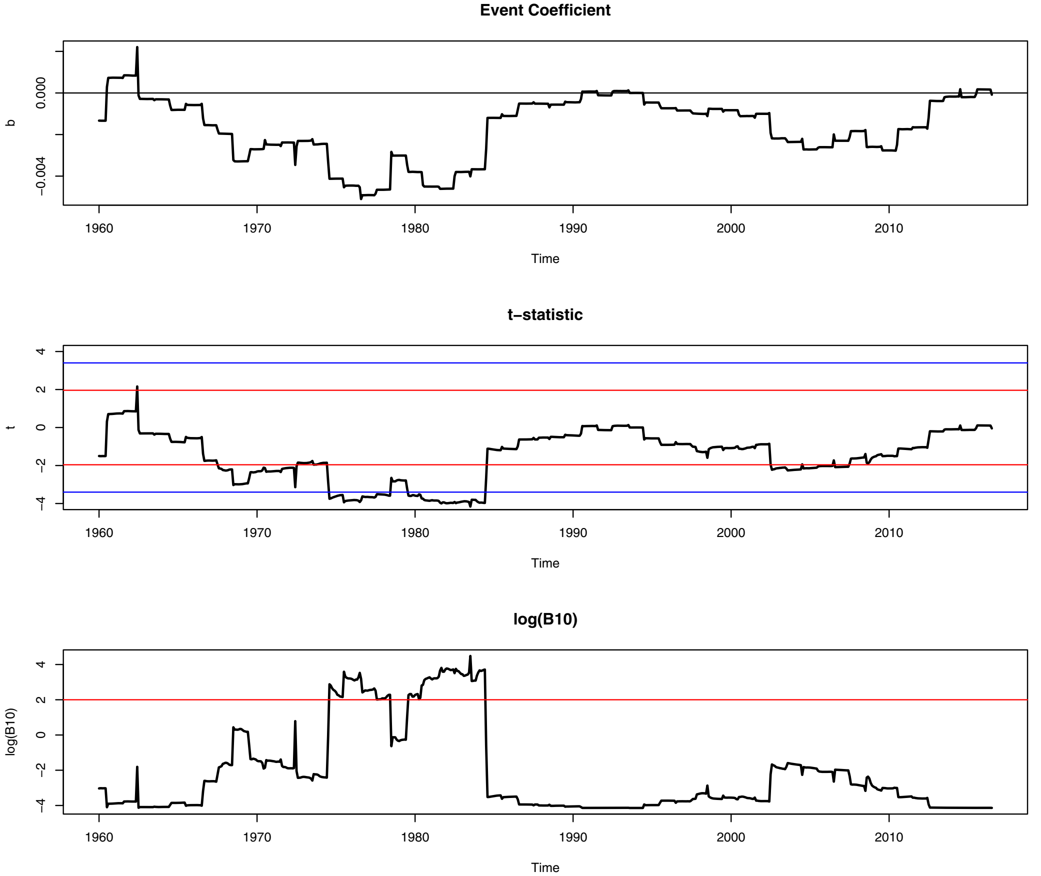

Figure 9. FIFA Effect Over Time with10-Year Rolling Window. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: For t-statistics, the red horizontal lines represent the two-tailed critical value at the 4% level of significance; and the blue lines represent the two-tailed critical value implied by the adaptive level of significance given in (11). For the $log(B_{10})$, the red line corresponds to 2, which is the threshold for 'positive' evidence.

where $\frac{I_{it}}{K_{it-1}}$ is firm i's investment rate for time t, σ_{it} is the firm ith idiosyncratic risk at time t, Z_{it} is the vector of controls (including firm's Tobin's Q, cash flow, leverage and size K), while η_i and g_t indicate fixed effects for firm and time. The data set includes 104,646 firm-year observations for all publicly traded firms in Compustat from 1970 to 2005; Panousi and Papanikolaou (2012) provide more data details. The key parameter of interest is γ_1, the coefficient of $log(\sigma_{it-1})$. Panousi and Papanikolaou (2012) hypothesize a negative relationship with $\gamma_1 < 0$. The relationship in (17) is of linear-log form, which means that a 1% increase of σ_{it-1} is expected to change $\gamma_1/100$ unit of the investment rate; and the elasticity of the investment rate with respect to σ_{it-1} is γ_1/y, where y is the dependent variable $\frac{I_{it}}{K_{it-1}}$ (see Carter Hill *et al.*, 2011; p.143).

4.3.1 *Replication and Effect Size*

Using the data provided by the authors,[2] I have almost exactly replicated their benchmark results reported in their table 1, using *R* (R Core Team, 2017). The estimated value of γ_1 from their benchmark model is -0.0196 with the *t*-statistic of -9.00 using the standard error clustered at the firm level. While the observed *t*-statistic of -9.00 indicates a clear rejection of $H_0 : \gamma_1 = 0$, the incremental contribution of $log(\sigma_{it-1})$ measured by $R_1^2 - R_0^2$ is only 0.0009, indicating that the signal-to-noise ratio is practically zero. As apparent from the expression given in (2), the test statistic is unable to reveal the information about the tiny incremental contribution of the variable of interest, since it is swamped by a massive sample size. The estimated coefficient value of -0.0196 indicates that a 1% increase of σ_{it-1} is expected to decrease the investment rate by around 0.0002 units. For firms with the value of investment rate ranging from 0.1 to 0.2, the elasticity estimate is in between -0.1 and -0.2, meaning that a 1% increase of σ_{it-1} is expected to decrease the investment rate by 0.1–0.2%.

4.3.2 *Alternatives to the p-Value Criterion*

The Bayes factors (not reported) also strongly favour H_1, while the adaptive level of significance (α^+) and the optimal level (α^*) are practically 0. Both P_{H_0} values based on (9) and (10) are practically 0. These results indicate that H_0 is highly unlikely to hold, which is consistent with the result based on the *p*-value criterion. This is not surprising given the massive sample size ($T = 104, 646$), in view of the zero probability paradox and the associated large-sample problems (Rao and Lovric, 2016), as discussed in Section 2. This is also a reflection of the property that any consistent test (classical or Bayesian) almost surely rejects H_0 with large or massive sample size, as discussed in Section 3. While these results indicate that H_0 is violated, this does not necessarily mean that σ_{it-1} has an economically significant impact on the investment rate, with reference to the fallacy of rejection (Spanos, 2017) discussed in Section 3. That is, given the massive sample size, it is important to note that the *t*-statistic of -9.00 is not due to a large effect size clearly different from 0, but due to a massive sample size. In light of these observations, it is important to conduct a range of descriptive and cautionary analyses, including sub-sample estimation of the model (17), in order to check whether the results are subject to large-sample bias in estimation and inference, as discussed in Section 2.

4.3.3 *Graphical Analyses*

To gain visual impressions of the relationship, I plot $\frac{I_{it}}{K_{it-1}}$ against $log(\sigma_{it-1})$, sorted based on the percentiles of $log(\sigma_{it-1})$ and the firms' Tobin's *Q*: see Figures B1 and B2 in the Appendix. It appears that the two variables visually show weak linear association in most cases, often giving the impression of a positive relationship. In many cases, a non-linear relationship looks likely. The linear regression lines are also plotted in red, which show no clear negative relationships in many cases. In addition, there are a large number of outliers which can distort the estimated relationship. Figure B3 presents the scatter plots for individual firms randomly selected among those appear for all 36 years, which provide similar visual impressions of no clear negative relationship. Overall, these visual representations indicate that: firstly, it is difficult to justify a negative linear relationship; and secondly, a single linear regression line falls well short of capturing the full facets of the data.

4.3.4 *Further Cautionary Analyses for Large-Sample Bias*

In relation to Figure B3, I also fit the regression model (17) individually to all 3841 firms appearing more than 10 times in the data set (without the fixed effect terms). From these regressions, the distribution

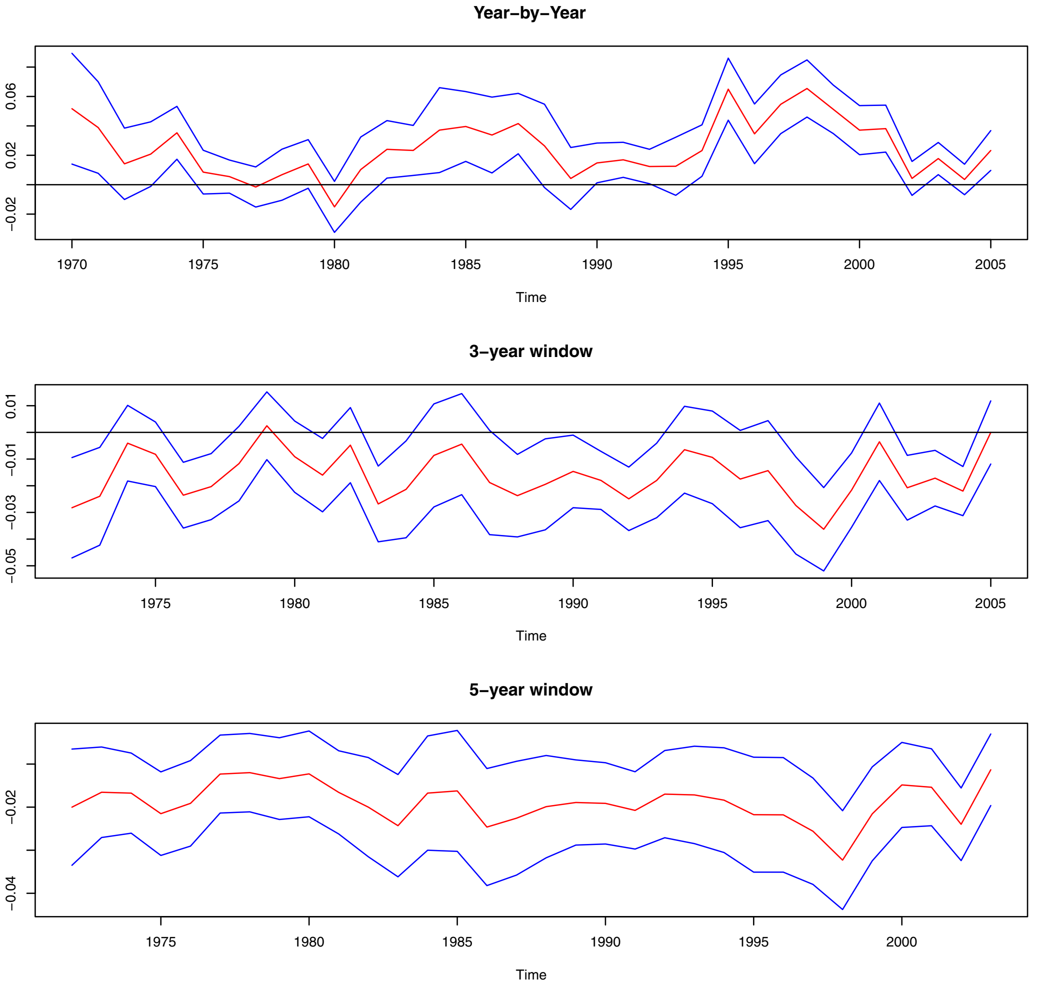

Figure 10. Sub-Sample Estimation of Model (17). [Colour figure can be viewed at wileyonlinelibrary.com]
Note: Each graph plots point estimates of γ_1 (in red) and their 95% confidence interval (in blue). Where applicable, firm-clustered standard error estimates are used.

of the estimates of γ_1 is examined. It has the mean of -0.014, the first quartile of -0.063, the second quartile (median) of -0.008 and the third quartile of 0.044. Around 45% of the estimates of γ_1 are found to be positive. The interval constructed based on the 2.5th and 97.5th percentiles of the distribution is $[-0.454, 0.383]$. These values show that the negative linear association between $\frac{I_{it}}{K_{it-1}}$ and $log(\sigma_{it-1})$ is not so strong at the individual firm level.

Figure 10 plots the estimates of γ_1 for the model (17), along with their 95% confidence intervals, when different sub-samples are considered. For year-by-year estimation (cross-sectional regression including lagged values), the coefficients are positive and statistically significant. For the 3-year overlapping window (fixed-effects model with firm-clustered standard errors), the estimates are mostly negative and statistically

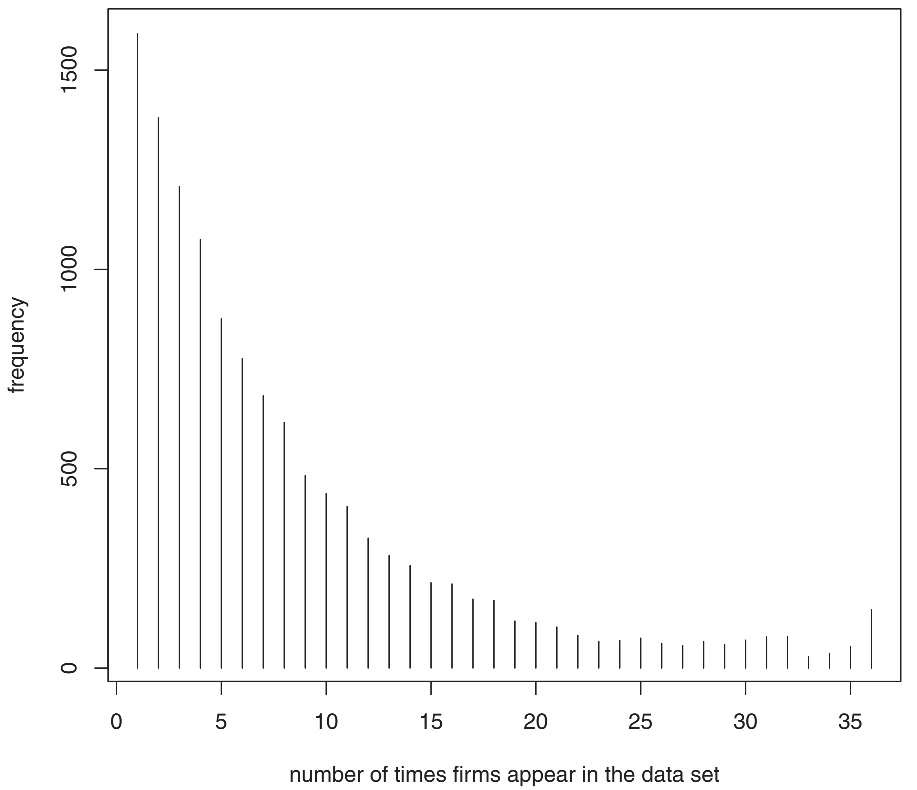

Figure 11. Frequency of the Firms.

insignificant; while for the 5-year window, the estimates are negative and statistically significant. It appears that, as the sample size increases, the negativity of the coefficient estimates and their statistical significance (at the 5% level) improve. This indicates that the main result of Panousi and Papanikolaou (2012) depends sensitively on the sample size.

Figure 11 plots the number of times firms appear in the data set. There are 1594 firms which appear only once in the data set. The number of firms that appear five times or less in the data set is 16,657, which makes up nearly 16% of the total observations. The number of firms-year observations that appear 30 times or more is 16,407, which makes up only 16% of the total observations. The presence of a large number of firms that appear in the data set only a small number of times means there are many data points which do not show much time series variations. In particular, the 1594 firms that appear only once in the data set have their within variations all exactly zero. A large number of zero values in a data set can bias estimation and statistical inference (see, for example, Correia, 2015). In addition, given the dynamic structure of (17), small time series variations in the data set can also bias the estimation results.

To examine the impact of these observations, I progressively remove the observations with a small number of appearances from the data set. Figure 12 plots the estimates of γ_1 from model (17) along with their 95% confidence intervals. The first points are associated with the model with the whole data set (including the firms appearing at least once); the next points are associated with the data set including the firms appearing at least twice; and so on. This continues until the observations for the firms that appear

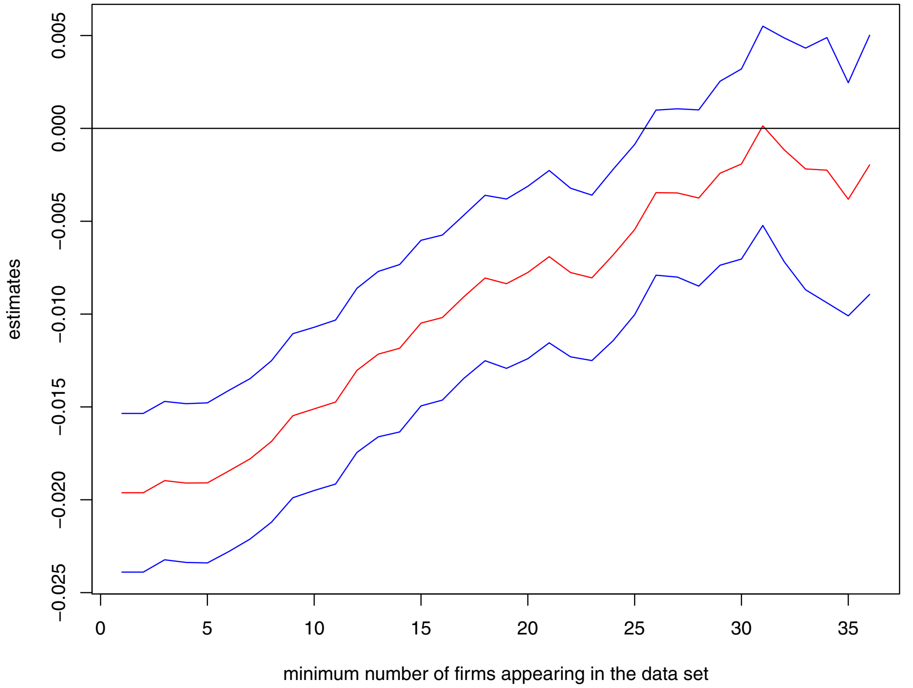

Figure 12. Sub-Sample Estimation of Model (17). [Colour figure can be viewed at wileyonlinelibrary.com]
Note: Estimation with progressively removing the firms that appear with a small frequency. Starting with the data set where firms appear at least once (whole sample), at least twice and so on; until only the firms appearing for all periods (36 years) are included. The point estimates of γ_1 (in red) and their 95% confidence interval (in blue) are plotted. Firm-clustered standard error estimates are used.

in all years (146 firms). As the observations are removed, the estimates converge to 0 and the confidence intervals cover zero indicating statistical insignificance. This indicates that the bias of model estimation and inference increases as more observations are added to the data set, which may be the effect of selection bias as these observations may well not be random. This may represent the case where estimation bias is being magnified as the sample size increases, which can be a problem in large-sample analysis (Kaplan *et al.*, 2014).

As well, there are a large number of firms whose time series observations appear with discontinuity. For example, the firm with the ID 76254 appears only twice in 1992 and 2003; the firm with ID 56573 has time series observation from 1975 to 1987 and then from 1998 to 2003. The number of firms with at least 1 year of discontinuation is 1581; at least 5 years of discontinuation is 358; and 63 firms have more than 10 years of discontinuation. These observations can be regarded as missing observations, meaning they can bias the estimation and testing results.

The discussions presented in sub-section suggest that there are elements of systematic bias in the data set, which can adversely affect the estimation and statistical inference; and that this bias shows up with a stronger force as the sample size gets larger. These observations are consistent with the concerns raised by a number of authors such as Kaplan *et al.* (2014), Harford (2014), Hand (2016, 2018) and Fiebig (2017). Overall, the descriptive and cautionary analyses conducted in this section cast serious doubts on the statistically significant result of Panousi and Papanikolaou (2012) based on the p-value criterion.

5. Concluding Remarks

In many fields of science, statistical decisions are often made based solely on the *p*-value criterion. This practice has a number of serious shortcomings, leading to a large number of false positive findings (Ioannidis, 2005; Rao and Lovric, 2016). A number of authors have warned against this practice. In particular, the ASA states (Wasserstein and Lazar, 2016) that

> The widespread use of "statistical significance" (generally interpreted as "$p \leq 0.05$") as a license for making a claim of a scientific finding (or implied truth) leads to considerable distortion of the scientific process.

As Gigerenzer (2004) argues, this practice promotes 'statistical rituals' and 'mindless statistics', discouraging statistical thinking and the art of choosing a proper tool for a given problem. In many areas of business research, the *p*-value criterion is almost exclusively used, as Kim and Ji (2015) and Kim *et al.* (2018) report from their extensive surveys. It is particularly problematic with the availability of large data sets, since hypothesis testing using the *p*-value criterion can be severely biased towards Type I error in this case, frequently producing research outcomes which are false positives. This suggests that the researchers should adopt a range of sensible alternatives to the *p*-value criterion, in order to make more informed inferential decisions in their research.

Moreover, with large or massive data sets, the researchers tend to be over-confident about the quality of their data. While large data sets constitute a powerful tool and can open up new research opportunities, there are growing concerns that many researchers are not fully aware of their possible dangers, which can mislead their research outcomes (see, for example, Boyd and Crawford, 2011; Wigan and Clarke, 2013). These dangers include sampling error, selection bias, measurement error, missing information, aggregation error and neglect on effect size: see, for example, Kaplan *et al.* (2014), Harford (2014), Hand (2016, 2018) and Fiebig (2017). The researchers should also note that most of their inferential tools can be severely biased towards Type I error under a massive sample size (Weakliem, 2016), and their decisions in this case can often be subject to the fallacy of rejection (Spanos, 2017). This means that researchers with large data sets should conduct extensive descriptive, cautionary and sensitivity analyses to ensure that their data sets are not subject to such problems. This is because, when such problems exist, it is highly likely that estimation bias is magnified with increasing sample size (Kaplan *et al.*, 2014).

This paper makes a contribution to the practice of empirical research in business studies by proposing that researchers employ a toolbox containing a range of credible alternatives to the *p*-value criterion in their statistical decision, adding to the recent papers such as Kim and Ji (2015), Harvey *et al.* (2016), Harvey (2017) and Ohlson (2018). By making an informed decision using these alternatives, we can promote statistical thinking, instead of mindless statistical rituals, also avoiding false positive research outcomes. As mentioned in Section 3.8, the researchers can choose their own instruments depending on the problems at hand and the context of their research. The toolbox proposed in this paper contains: the optimal level of significance of Leamer (1978); equal-probability test of Arrow (1960) and Naik (1969); adaptive level of significance of Perez and Perrichi (2014); Bayes factor (Zellner and Siow, 1980; Liang *et al.*, 2008); approximate probability of the null hypothesis being true (Startz, 2014); and sample size selection. It also includes equivalence testing (Lauzon and Caffo, 2009; Rao and Lovric, 2016), as well as elementary data and graphical analyses to guard against potential large-sample bias. I have applied the toolbox proposed in this paper to three notable studies in finance, with the findings that their results based solely on the *p*-value criterion cannot stand under the alternative instruments in the toolbox.

It is alarming that many established empirical findings in finance and other business areas are obtained using the *p*-value criterion almost exclusively, often under a large or massive sample size. The results of this paper suggest that, in order to maintain research credibility and integrity at an elevated level, researchers change the ways in which they conduct statistical analysis and make statistical decisions.

They should employ more sensible alternatives to the *p*-value criterion in their inductive decision making, also conducting a range of basic data analyses to guard against possible large-sample bias in estimation and inference. This may require less dependance on inferential statistics, which have little economic significance. Instead, the researchers should focus more on the estimation-based measures such as the effect size (e.g. elasticity), model's explanatory power (e.g. R^2 and signal-to-noise ratio), confidence interval and transparent descriptive data analyses including skilful graphical methods (see Leamer, 1988; Gigerenzer, 2004; Soyer and Hogarth, 2012, Ohlson, 2018). More broadly, as Ohlson (2018) argues, researchers should change their research mentality that conforms with the 'generally accepted research procedures' or follows the 'rules of the game'. As Rao and Lovric (2016; p.14) assert, in the 21st century where researchers deal with large data sets and complex questions, a paradigm shift in testing statistical hypotheses is called for. It will indeed be a tall order to change the prevailing research culture and practice, but as Gigerenzer (2004) asserts,

> To stop the ritual, we also need more guts and nerves. We need some pounds of courage to cease playing along in this embarrassing game. This may cause friction with editors and colleagues, but it will in the end help them to enter the dawn of statistical thinking.

Notes

1. Harford (2014) provides several illuminating examples including the failure of Google Flu Trends and the debacle concerning the 1936 U.S. Presidential poll by Literary Digest.
2. I would like to thank the authors for kindly providing the data.

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Appendix A: Box-and-Whisker Plots

This section presents the Box-and-Whisker plots of the data for political cycle and stock return (Section 3.1) and for the FIFA World Cup effect on stock return (Section 3.2).

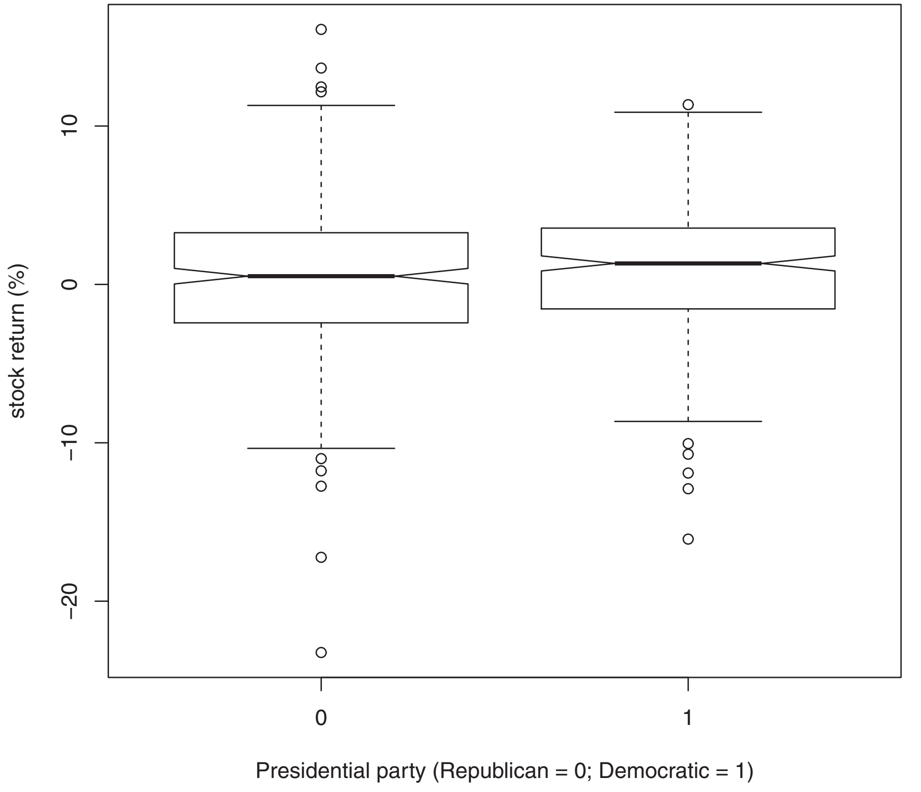

Figure A1. Stock Return and Political Cycle.
Note: Notched Box-and-Whisker plots are presented. If the notches of the two boxes overlap, the medians are statistically not different (see McGill *et al.*, 1978).

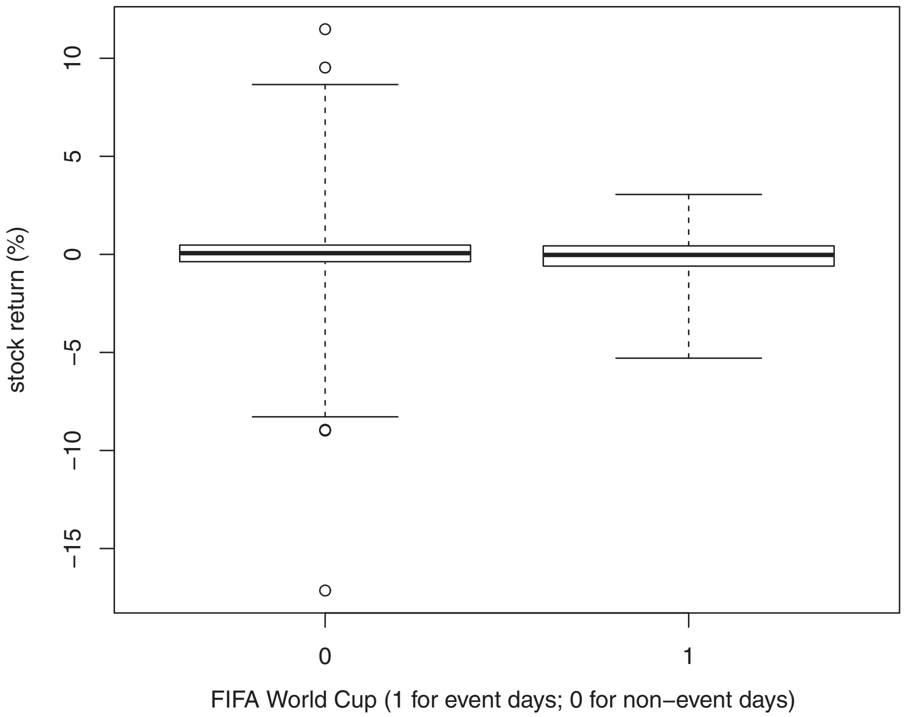

Figure A2. Stock Return and FIFA World Cup.

Appendix B: Scatter Plots

This section visually presents the data used by Panousi and Papanikolaou (2012), discussed in Section 3.3. Figures B1 and B2 show the plots (smoothed colour density representation) of $\frac{I_{it}}{K_{it-1}}$ (Y-axis) against $log(\sigma_{it-1})$ (X-axis), sorted by $log(\sigma_{it-1})$ and Tobin's Q ($log(Q_{it-1})$). They are classified into nine subsets based on 10 equal-spaced percentiles (0th, 11th, 22th, ..., 89th, 100th). The first graph plots the data in the first set ([0th, 11th] percentiles), the second plots the data in the second set ([11th, 22th] percentiles) and so on.

The background of Figure B3 shows the scatter plot (smoothed colour density representation) of $\frac{I_{it}}{K_{it-1}}$ (Y-axis) against $log(\sigma_{it-1})$ (X-axis), for the 146 firms that appear for the entire period of 36 years. In each graph, the red, blue, yellow and green dots indicate the points associated with the firms randomly selected from these 146 firms.

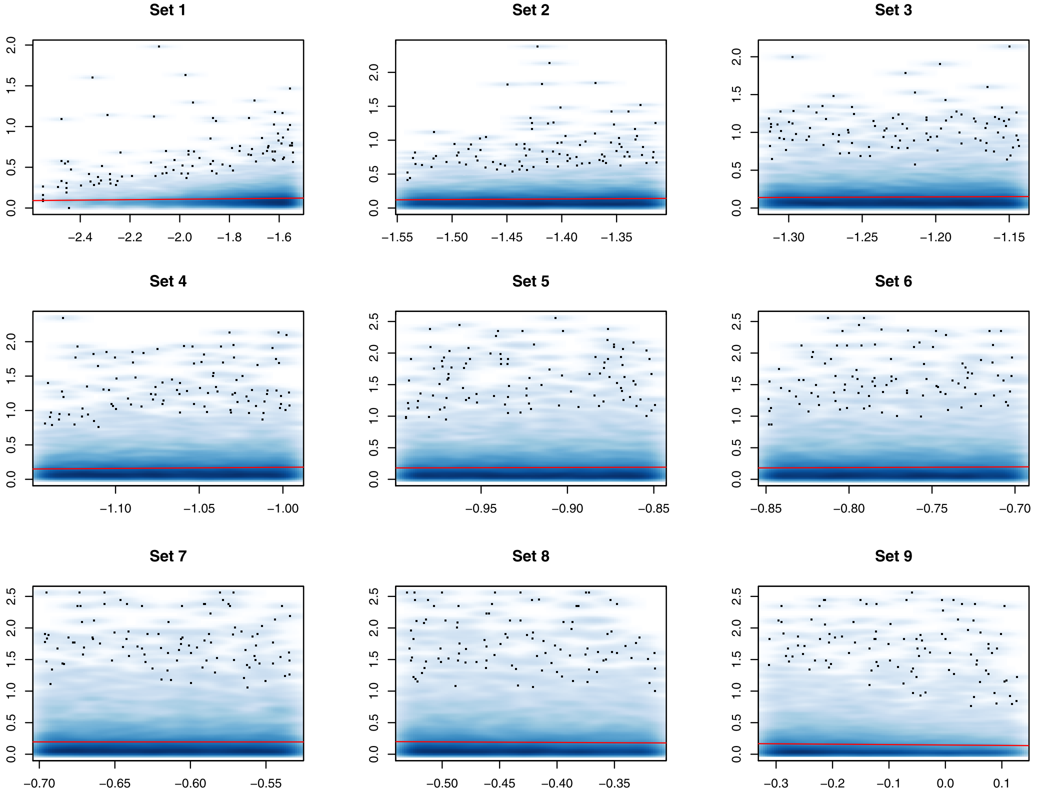

Figure B1. Plots of $\frac{I_{it}}{K_{it-1}}$ Against $log(\sigma_{it-1})$ Sorted by $log(\sigma_{it-1})$. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The red line represents the fitted linear regression line.

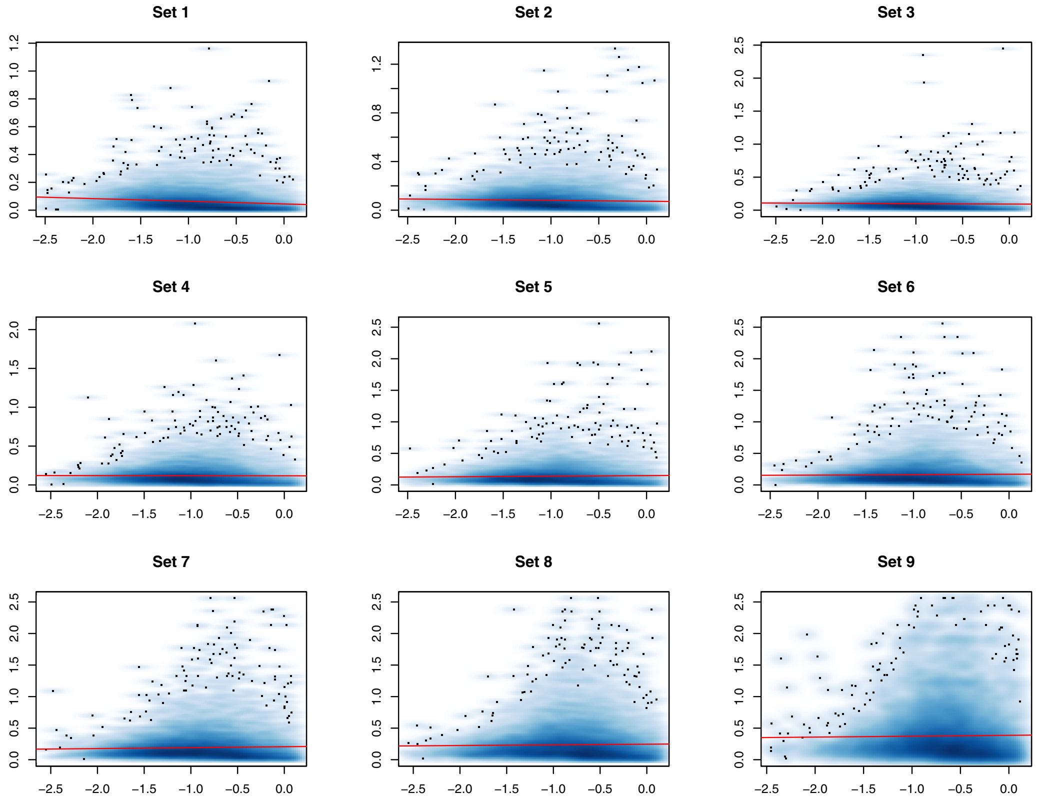

Figure B2. Plots of $\frac{I_{it}}{K_{it-1}}$ Against $log(\sigma_{it-1})$ Sorted by $log(Q_{it-1})$. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: The red line represents the fitted linear regression line. Q represents Tobin's Q.

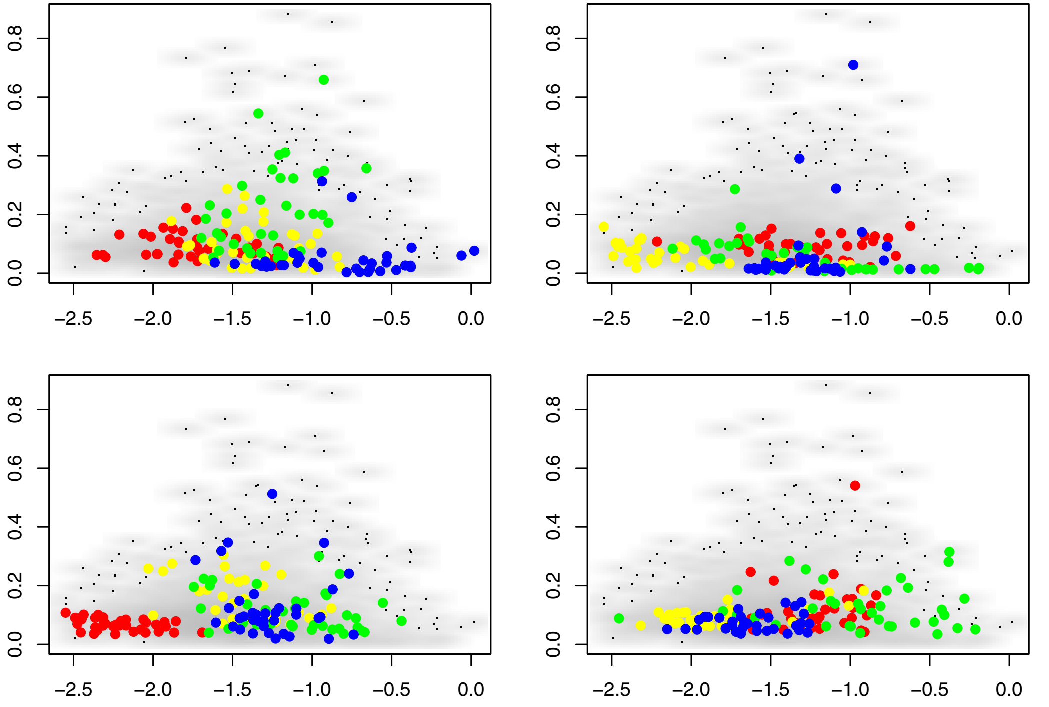

Figure B3. Plots of $\frac{I_{it}}{K_{it-1}}$ Against $log(\sigma_{it-1})$ for Selected Firms. [Colour figure can be viewed at wileyonlinelibrary.com]

Note: For each graph, the grey background represents the scatter for the 146 firms appearing for the entire period of 36 years. The dots of a particular colour match the points of a firm randomly selected from these firms.